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(54) **BIVALENT BISPECIFIC ANTIBODY AND PREPARATION METHOD THEREOF, CODING GENE, HOST CELL AND COMPOSITION**

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None

See application file for complete search history.

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ABSTRACT

Provided are a bivalent bispecific antibody and a preparation method thereof, a coding gene, a host cell and a composition. The bivalent bispecific antibody comprises: a) a single-chain variable fragment scFv, a flexible peptide, a heavy chain IgG1 constant region CH1 and a hinge region partial sequence of the antibody that specifically binds to a first antigen, and b) a single-chain variable fragment scFv, and a light chain constant region CL, that is, scFv1-CL or CL-scFv1, of the antibody that specifically binds to a second antigen; or comprises: c) a light chain and a heavy chain of the antibody that specifically binds to the first antigen, and d) a light chain and a heavy chain of the antibody that specifically binds to the second antigen.

8 Claims, 4 Drawing Sheets

Specification includes a Sequence Listing.

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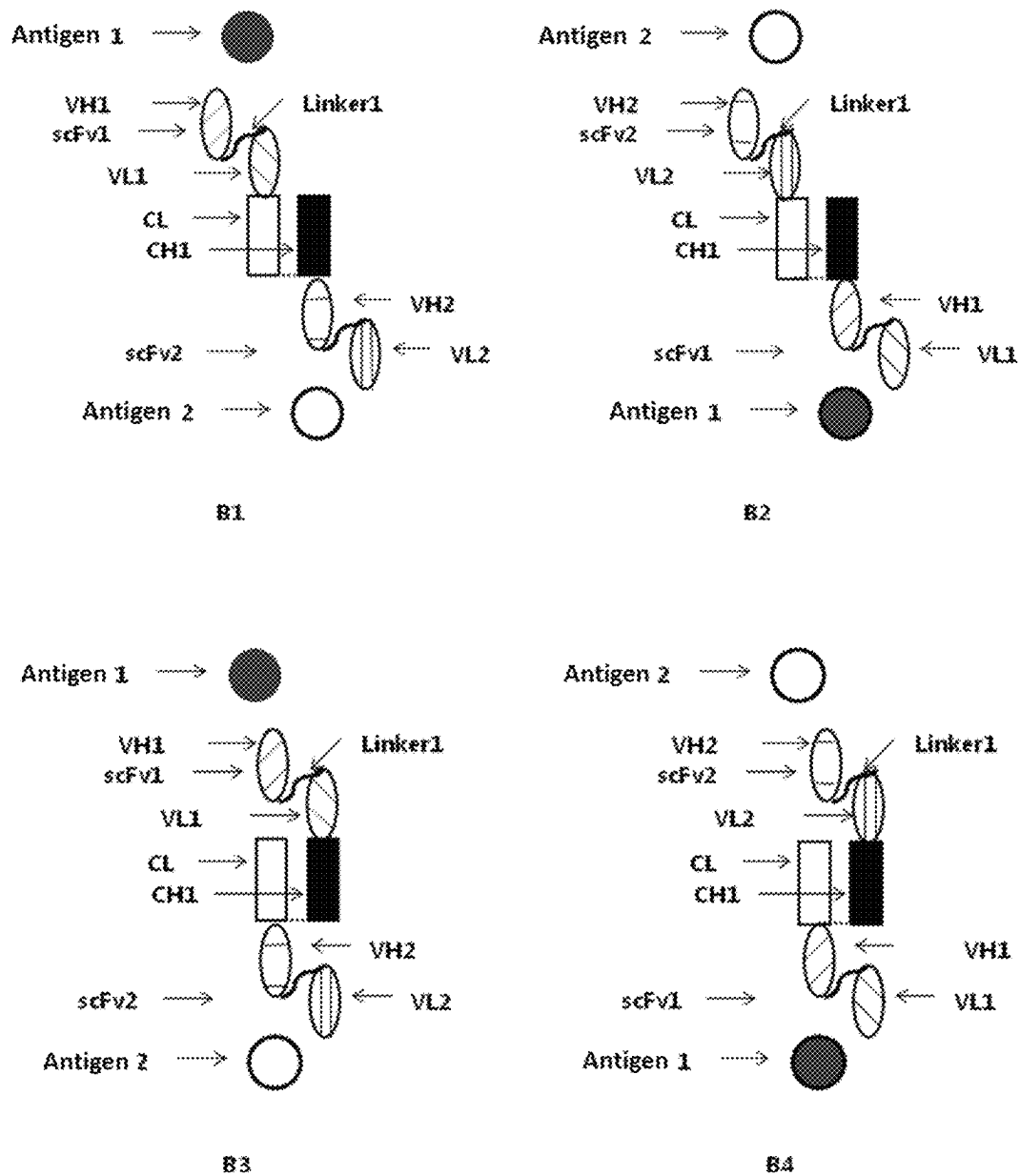


FIG. 1

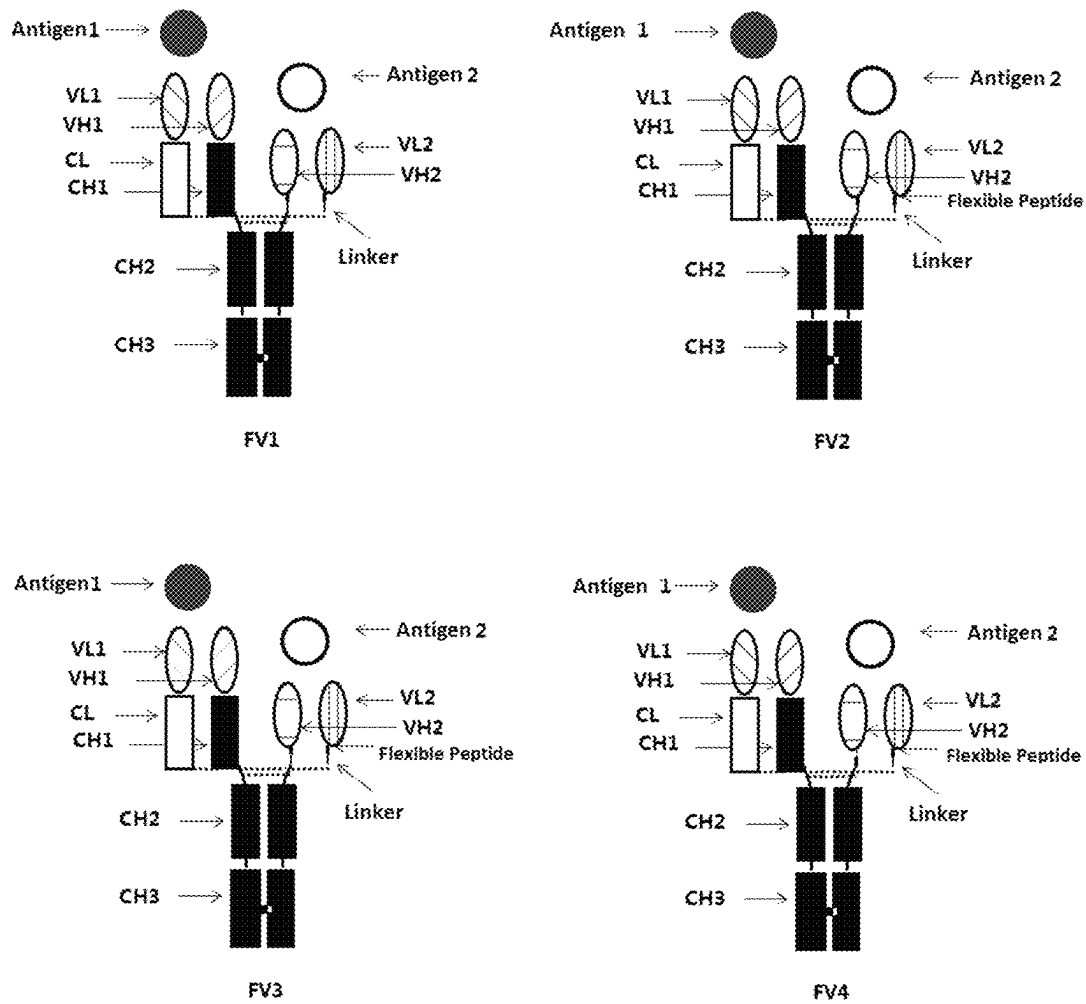


FIG. 2

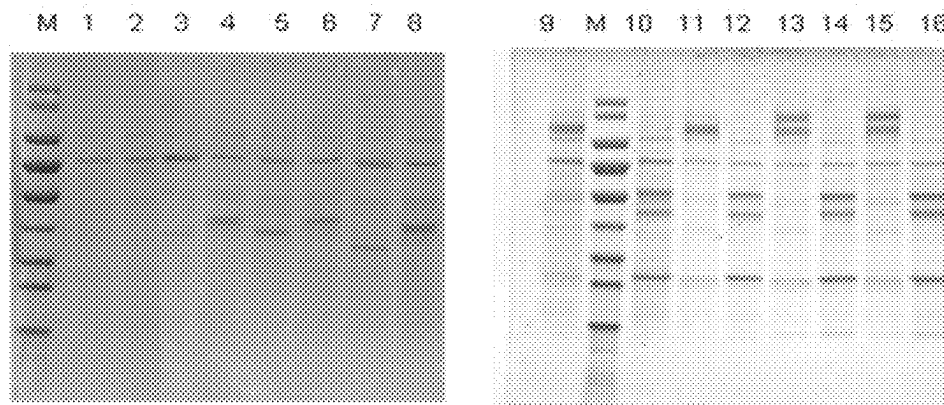


FIG. 3

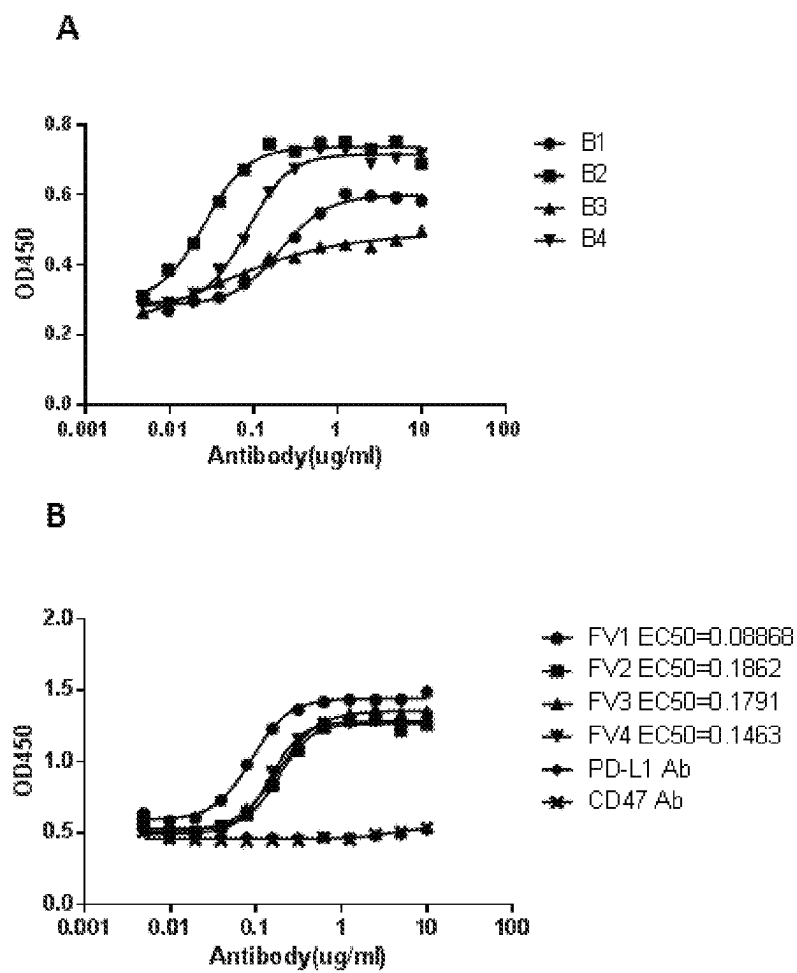


FIG. 4

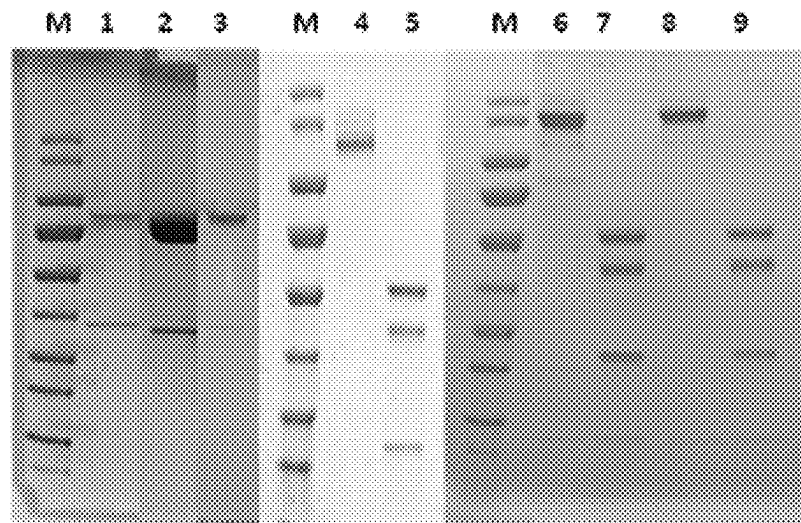


FIG. 5

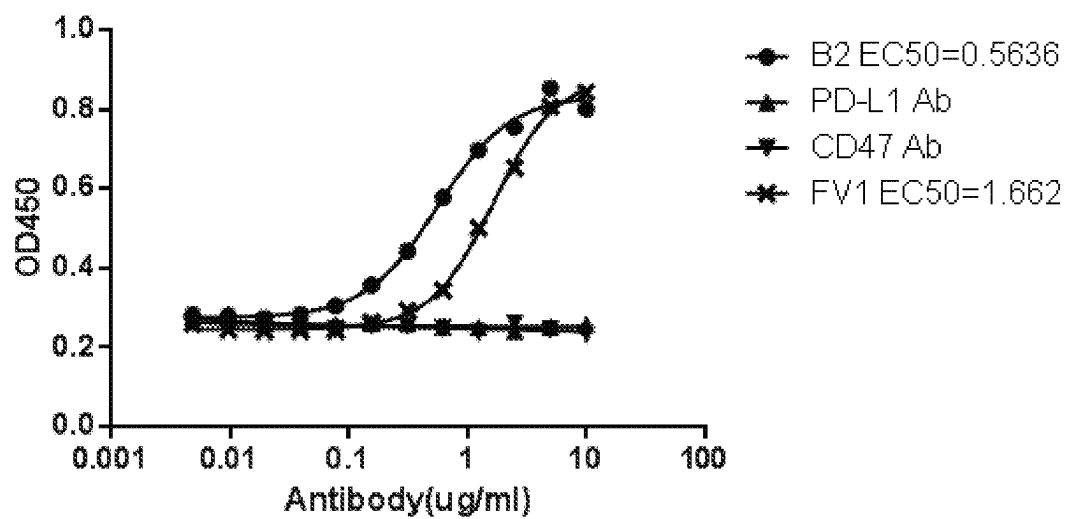


FIG. 6

1

BIVALENT BISPECIFIC ANTIBODY AND PREPARATION METHOD THEREOF, CODING GENE, HOST CELL AND COMPOSITION

CROSS REFERENCE TO RELATED APPLICATIONS

This application is a US National Phase application based upon PCT Application No. PCT/CN2019/128582, filed Dec. 26, 2019, which claims the priority of Chinese Patent Application No. 201811622069.2, filed to China National Intellectual Property Administration on Dec. 28, 2018, and titled with “BIVALENT BISPECIFIC ANTIBODY AND PREPARATION METHOD THEREOF, CODING GENE, HOST CELL AND COMPOSITION”, and the disclosures of which are hereby incorporated by reference.

SEQUENCE LISTING

This application contains references to amino acid sequences and/or nucleic acid sequences which have been submitted concurrently herewith as the sequence listing text file entitled “69427904_210033-APXU-GENSCI-Sequence_listing-v.”, file size S6,643 bytes, created on Mar. 5, 2025. The aforementioned sequence listing is hereby incorporated by reference in its entirety.

FIELD

The present disclosure relates to the technical field of antibody medicine, specifically to a bivalent bispecific antibody and preparation method thereof, coding gene, host cell and composition.

BACKGROUND

A bispecific monoclonal antibody (BsAb) is a special antibody that is artificially made to bind two different antigens at the same time. Bispecific antibodies can recognize both tumor target cells and immune effector cells, so they have dual functions of antibody specificity and mediating the cytotoxicity of effector cells. Bispecific antibodies can recruit effector cells at tumor sites and activate effector cells to exert anti-tumor effects. The mechanism of killing tumor cells mediated by bispecific antibodies includes cell proliferation, cytokine release, cytotoxic peptides and regulation of enzymes. In vivo and clinical studies have proved that bispecific antibody-mediated immunotherapy can treat tumors in some animals, and clinically can mitigate the condition of patients with tumor and prolong their life. Therefore, the application of bispecific antibody-mediated immunocompetent cells in tumor therapy has a good prospect.

Bispecific antibodies are not nature products and can only be prepared artificially. Bi- or multi-specific antibodies in the art can bind to at least two antigens and can be produced by cell fusion, chemical conjugation or recombinant DNA technology. Recently, a wide variety of recombinant bispecific antibody structures have been developed, such as tetravalent bispecific antibodies by fusion of, for example, an IgG antibody and a single-chain domain (Coloma, M. J., et al., *Nature Biotech.* 15 (1997) 159-163; WO2001077342; and Morrison, S. L., *Nature Biotech.* 25 (2007) 1233-1234). In addition, many other new forms that can bind to more than two antigens have been developed, in which the main structure of the antibody (IgA, IgD, IgE, IgG or IgM) is no

2

longer limited to, such as diabodies, triabodies or tetrabodies, minibodies and several single-chain forms (scFv, Bis-scFv) (Holliger, P., et al., *Nature Biotech.* 23 (2005) 1126-1136; Fischer, N., and Léger, O., *Pathobiology* 74 (2007) 3-14; Shen, J., et al., *Journal of Immunological Methods* 318 (2007) 65-74; Wu, C., et al., *Nature Biotech.* 25 (2007) 1290-1297).

In one method, the cell quadroma technology (Milstein, C. and A. C. Cuello, *Nature*, 305 (1983) 537-40) is utilized to produce a bispecific antibody that is very similar to a natural antibody. The cell quadroma technology is based on the somatic fusion of two different hybridoma cell lines expressing murine monoclonal antibodies with the desired bispecific antibody specificity. Because of the random pairing of two different heavy and light chains of antibodies in the hybridoma cell lines, up to 10 different antibody types will be generated, of which only one is the desired functional bispecific antibody. Due to the presence of mismatched by-products and significantly low yields, complicated purification procedures are required (Morrison, S. L., *Nature Biotech* 25 (2007) 1233-1234). Generally, if recombinant expression technology is used, the same problem of mismatch by-products still exists.

A method used to avoid the problem of mismatch by-products is called “knobs-into-holes”. The purpose is to force the heavy chains from two different antibodies to pair with each other by introducing mutation into the CH3 domain to modify the contact interface. In one chain, amino acids with large volume are replaced by amino acids with short side chains to form a “hole”. Conversely, amino acids with a large side chain are introduced to the other CH3 domain to form a “knob”. By co-expressing these two heavy chains, a higher yield of heterodimer form (“knob-hole”) compared with homodimer form (“hole-hole” or “knob-knob”) was observed (Ridgway, J. B., Presta, L. G., Carter, P. and WO 1996027011). The percentage of heterodimer can be further increased by reconstruction of the interaction interface of the two CH3 domains using phage display method and introduction of disulfide bonds to stabilize the heterodimer (Merchant, A. M., et al., *Nature Biotech* 16 (1998) 677-681; Atwell, S., Ridgway, J. B., Wells, J. A., Carter, P., *J. Mol. Biol.* 270 (1997) 26-35). An important constraint of this strategy is that the light chains of the two parent antibodies must be the same to prevent mismatches and formation of inactive molecules.

In addition to the “knob-hole” structure, the Fc pairing of different half-antibodies can also be achieved through the strand-exchange engineered domain (SEED) technology of IgG and IgA CH3 (Davis, J. H., et al., *Protein Eng. Des. Sel.*, 2010, 23 (4): 195-202).

In order to solve the problem of the incorrect assembly of different light chains, a new process of double-cell line expressing half-antibodies separately and in vitro assembly has been developed recently. Inspired by the half-antibody random exchange process of human IgG4 antibodies naturally occurring under physiological conditions, GenMab has developed FAE (Fab-arm exchange) bifunctional antibody technology (Gramer, M. J., et al., *MAbs* 2013, 5 (6): 962-973.). Introducing two point mutations, K409R and F405L, into the IgG1 heavy chain CH3 domains of the two target antibodies respectively, can produce half-antibody exchange rearrangement similar to that of IgG4 antibodies. Two different IgG1 antibodies after mutation were expressed in two CHO cell lines respectively, and the assembly between the light and heavy chains of each half-antibody was completed. After protein A affinity purification, precise

assembly between heterogeneous half-antibodies can be achieved in vitro by using a mild oxidant system.

In addition to sharing light chains with the same sequence or performing in vitro assembly, the correct assembly of light chains of antibodies can also be facilitated by Crossmab technology. A representative product is Roche's Ang-2/VEGF CrossMab CH1-CL. Based on the modification of "knobs-into-holes", Crossmab technology exchanged CL and CH1 in the Fab domain of Ang-2 antibody and remained the Fab domain of VEGF antibody unchanged. The light chain of the modified Ang-2 antibody is not easily mismatched with the heavy chain of the VEGF antibody, and the "knob-hole" structure can promote the heterodimerization of the two heavy chains (Schaefer, W, et al., Proc Natl. Acad. Sci. USA, 2011, 108 (27): 11187-11192).

Moreover, two single-chain antibodies (scFv) or two Fabs can be linked through a peptide to form a bifunctional antibody fragment. A representative product is BiTE (bispecific T-cell engager) series products developed by Micromet in German. This series of products is generated by linking anti-CD3 single-chain antibodies with the single-chain antibodies against different anti-tumor cell surface antigens through a peptide (Baeuerle, P. A., et al., Cancer Res., 2009, 69 (12): 4941-4944). The advantage of such antibody structure is that it has a small molecular weight, can be expressed in prokaryotic cells, and does not require the consideration of incorrect assembly; while the disadvantage is that it cannot mediate some corresponding biological functions due to a lack of antibody Fc fragment, and its half-life is short.

The patent application publications US2015/0284475A1 and CN101896504A disclose bivalent bispecific antibodies, but the bivalent bispecific antibodies in the two applications have low affinity for antigens. The flexible peptide disclosed in "Prediction of three dimensional structure of bispecific antibodies linked by flexible peptide" published by Shao Changli, was used to modify the bivalent bispecific antibodies in the two aforementioned applications, but the affinity of the antibodies was still not ideal.

In view of the problems of light chain mismatches, low correct assembly rate of light and heavy chains, and the too large or small molecule size of the above-mentioned bispecific antibodies, it is necessary to develop a new type of a bivalent bispecific antibody.

SUMMARY

In view of this, the present disclosure provides a bivalent bispecific antibody and a preparation method thereof, a coding gene, a host cell and a composition. The antibody has a less mismatched assembly rate of light and heavy chains and a moderate molecular size.

In order to achieve the above objects of the present disclosure, the present disclosure provides the following technical solutions:

The present disclosure provides a bivalent bispecific antibody, comprising

- a) a single-chain variable fragment (scFv) derived from an antibody that specifically binds to a first antigen, a flexible peptide, a heavy chain constant region CH1 and a partial hinge region of IgG1, namely CH1-partial hinge-linker (flexible peptide)-scFv2 or scFv2-linker-CH1-partial hinge, and
- b) a single-chain variable fragment (scFv) derived from an antibody that specifically binds to a second antigen and a light chain constant region CL, namely scFv1-CL or CL-scFv1;

or comprising

- c) a light chain and a heavy chain of an antibody that specifically binds to a first antigen, and
- d) a light chain and a heavy chain of an antibody that specifically binds to a second antigen, wherein light chain variable region of the light chain is linked to a flexible peptide and a linker, and heavy chain variable region of the heavy chain is linked to a heavy chain Fc fragment through a flexible peptide and a linker.

Preferably, the flexible peptide comprises a sequence of (G4S/G4SAS) n, wherein n is an integer greater than or equal to 0, and the IgG1 partial hinge region linked to the flexible peptide comprises a sequence of EPKSCDK (SEQ ID NO: 24), wherein (G4S/G4SAS) n represents (GGGGS (SEQ ID NO: 25)) n_or (GGGGSAS (SEQ ID NO: 26)) n, and n is an integer greater than or equal to 0.

The linker comprises a sequence of L/GGGC (L/GGGC represents LGGC (SEQ ID NO: 27) or GGGC (SEQ ID NO: 28)), and the first cysteine residue (C) in heavy chain hinge region linked to the linker is mutated to serine(S).

Preferably, in steps a) and b), CL and CH1 form a heterodimer via a disulfide bond between the terminal cysteine residue in CL and the cysteine residue in the heavy chain hinge region.

Preferably, in steps c) and d), the terminal cysteine residue in the linker LGGC (SEQ ID NO: 27) or GGGC (SEQ ID NO: 28) linked to the heavy chain and the terminal cysteine residue in the linker LGGC (SEQ ID NO: 27) or GGGC (SEQ ID NO: 28) linked to the light chain form a disulfide bond; and the CH3 domain of first heavy chain and the CH3 domain of second heavy chain are modified to a structure that facilitates the formation of the bivalent bispecific antibody.

Preferably, the modification comprises

- a) modification to the CH3 domain of the first heavy chain: in the interface between the CH3 domain of the first heavy chain and the CH3 domain of the second heavy chain of the bivalent bispecific antibody, an amino acid residue in the CH3 domain of the first heavy chain is replaced with an amino acid residue with a volume larger than the original amino acid residue to form a knob in the CH3 domain of the first heavy chain, wherein the knob is capable of inserting into a hole of the CH3 domain of the second heavy chain, and
- b) modification to the CH3 domain of the second heavy chain: in the interface between the CH3 domain of the second heavy chain and the CH3 domain of the first heavy chain of the bivalent bispecific antibody, an amino acid residue in the CH3 domain of the second heavy chain is replaced with an amino acid residue with a volume smaller than the original amino acid residue to form a hole in the CH3 domain of the second heavy chain, wherein the hole is capable of holding the knob of the CH3 domain of the first heavy chain;

wherein each of the first heavy chain variable region and the light chain variable region is linked to (GGGGS (SEQ ID NO: 25)) n_or (GGGGSAS (SEQ ID NO: 26)) n and LGGC (SEQ ID NO: 27) or GGGC (SEQ ID NO: 28), and a disulfide bond is formed between the cysteine residues of the two L/GGGC.

Preferably, the amino acid residue with a volume larger than the original amino acid residue is selected from the group consisting of arginine (R), phenylalanine (P), tyrosine (Y), and tryptophan (W).

5

The amino acid residue with a volume smaller than the original amino acid residue is selected from the group consisting of alanine (A), serine(S), threonine (T), and valine (V).

Preferably, the bivalent bispecific antibody provided by the present disclosure comprises:

- a) a single-chain variable fragment (scFv), a flexible peptide, a heavy chain constant region CH1 and a partial hinge region of IgG1, derived from an antibody that specifically binds to a first antigen, i.e., the amino acid sequence as shown in SEQ ID NO: 2, and
- b) a single-chain variable fragment (scFv) and a light chain constant region CL derived from an antibody that specifically binds to a second antigen, i.e., the amino acid sequence as shown in SEQ ID NO: 11;
- c) a light chain and a heavy chain of an antibody that specifically binds to a first antigen, i.e., the amino acid sequences as shown in SEQ ID NO: 5 and SEQ ID NO: 14, and
- d) a light chain and a heavy chain of an antibody that specifically binds to a second antigen, i.e., the amino acid sequences as shown in SEQ ID NO: 6 and SEQ ID NO: 15, wherein the light chain variable region is linked to a flexible peptide and a linker, and the heavy chain variable region is linked to the heavy chain Fc fragment through a flexible peptide and a linker.

The present disclosure also provides a method for producing the bivalent bispecific antibody, comprising the following steps:

- a) Transforming a host cell with the following vectors: a first vector (comprising the gene encoding SEQ ID NO: 2), comprising a nucleic acid encoding a single-chain variable fragment and a heavy chain constant region CH1 of an antibody that specifically binds to a first antigen, and
- a second vector (comprising the gene encoding SEQ ID NO: 11), comprising a nucleic acid encoding a single-chain variable fragment and a light chain constant region of an antibody that specifically binds to a second antigen;
- or with the following vectors:
- a third vector (comprising the gene encoding SEQ ID NO: 14), comprising a nucleic acid encoding a light chain of an antibody that specifically binds to a first antigen,
- a fourth vector (comprising the gene encoding SEQ ID NO: 5), comprising a nucleic acid encoding a heavy chain of an antibody that specifically binds to a first antigen,
- a fifth vector (comprising the gene encoding SEQ ID NO: 15), comprising a nucleic acid encoding a light chain of an antibody that specifically binds to a second antigen, wherein the light chain variable region is linked to a linker, and
- a sixth vector (comprising the gene encoding SEQ ID NO: 6), comprising a nucleic acid encoding a heavy chain of an antibody that specifically binds to a second antigen, wherein the heavy chain variable region is linked to the heavy chain Fc fragment through a linker;
- b) culturing the host cell under conditions that allow the synthesis of an antibody; and
- c) recovering the antibody from the culture.

The present disclosure also provides a nucleic acid molecule encoding the bivalent bispecific antibody of the present disclosure, comprising

- a first nucleic acid (the gene encoding SEQ ID NO: 2), encoding a single-chain variable fragment (scfv), a

6

flexible peptide, a heavy chain igg1 constant region CH1, and a partial hinge region of an antibody that specifically binds to a first antigen, and

- a second nucleic acid (the gene encoding SEQ ID NO: 11), encoding a single-chain variable fragment (scfv) and a light chain constant region CL of an antibody that specifically binds to a second antigen;

or comprising

- a third nucleic acid (the gene encoding SEQ ID NO: 14), encoding a light chain of an antibody that specifically binds to a first antigen,
- a fourth nucleic acid (the gene encoding SEQ ID NO: 5), encoding a heavy chain of an antibody that specifically binds to a first antigen,
- a fifth nucleic acid (the gene encoding SEQ ID NO: 15), encoding a light chain of an antibody that specifically binds to a second antigen, wherein the light chain variable region is linked to a flexible peptide and a linker, and
- a sixth nucleic acid (the gene encoding SEQ ID NO: 6), encoding a heavy chain of an antibody that specifically binds to a second antigen, wherein the heavy chain variable region is linked to the heavy chain Fc fragment through a flexible peptide and a linker.

The present disclosure also provides a host cell, comprising

- a first vector, comprising a nucleic acid encoding a single-chain variable fragment and a heavy chain constant region CH1 of an antibody that specifically binds to a first antigen, and
- a second vector, comprising a nucleic acid encoding a single-chain variable fragment and a light chain constant region of an antibody that specifically binds to a second antigen;
- or comprising
- a third vector, comprising a nucleic acid encoding a light chain of an antibody that specifically binds to a first antigen,
- a fourth vector, comprising a nucleic acid encoding a heavy chain of an antibody that specifically binds to a first antigen,
- a fifth vector, comprising a nucleic acid encoding a light chain of an antibody that specifically binds to a second antigen, wherein the light chain variable region is linked to a linker, and
- a sixth vector, comprising a nucleic acid encoding a heavy chain of an antibody that specifically binds to a second antigen, wherein the heavy chain variable region is linked to the heavy chain Fc fragment through a linker.

The present disclosure also provides a composition comprising the bivalent bispecific antibody, and the composition is a pharmaceutical composition or a diagnostic composition.

Preferably, the pharmaceutical composition also comprises at least one pharmaceutical excipient.

The present disclosure provides a bivalent bispecific antibody and a preparation method thereof, a coding gene, a host cell and a composition. The bivalent bispecific antibody comprises a) a single-chain variable fragment (scFv), a flexible peptide, a heavy chain constant region CH1 and a partial hinge region of IgG1, derived from an antibody that specifically binds to a first antigen, namely CH1-partial hinge-linker (flexible peptide)-scFv2 or scFv2-linker-CH1-partial hinge, and b) a single-chain variable fragment (scFv) and a light chain constant region CL derived from an antibody that specifically binds to a second antigen, namely scFv1-CL or CL-scFv1;

or comprises

c) a light chain and a heavy chain of an antibody that specifically binds to a first antigen, and d) a light chain and a heavy chain of an antibody that specifically binds to a second antigen, wherein light chain variable region of the light chain is linked to a flexible peptide and a linker, and heavy chain variable region of the heavy chain is linked to the heavy chain Fc fragment through a flexible peptide and a linker.

The present disclosure has the following technical effects.

The bivalent bispecific antibody of the present disclosure has high affinities to the first antigen molecule and the second antigen molecule, which are equivalent to the parent monoclonal antibody molecule. For example, the affinities of B2 and FV1 to PD-L1 are $1.09\text{E-}10$ M and $2.71\text{E-}10$ M respectively; and the affinities of B2 and FV1 to PD-L1 are $2.58\text{E-}8$ M and $1.50\text{E-}8$ M, respectively. It can be seen that the antibody of the present disclosure has a high correct assembly rate of light and heavy chains and a moderate molecular size.

BRIEF DESCRIPTION OF DRAWINGS

FIG. 1 is a schematic diagram of bivalent bispecific antibodies with B structure, including four structures: B1, B2, B3 and B4.

FIG. 2 is a schematic diagram of bivalent bispecific antibodies with FV structure, including four structures: FV1, FV2, FV3 and FV4. In the structure FV1, there is no flexible peptide; in the structure FV2, the sequence of the flexible peptide is GGGGS (SEQ ID NO: 25) or GGGGSAS (SEQ ID NO: 26); in the structure FV3, the sequence of the flexible peptide is GGGGSGGGGSGGGGS (SEQ ID NO: 29) or GGGGSASGGGGSASGGGGSAS (SEQ ID NO: 30); and in the structure FV4, the sequence of the flexible peptide is GGGGS (SEQ ID NO: 25) or GGGGSAS (SEQ ID NO: 26).

FIG. 3 shows the SDS-PAGE results of the transient expression of bispecific antibodies with B structure and with FV structure. M indicates Marker; Lanes 1, 3, 5, 7, 9, 11, 13, and 15 represent the non-reducing electrophoresis results of B1, B2, B3, B4, FV1, FV2, FV3, and FV4, respectively; and Lanes 2, 4, 6, 8, 10, 12, 14, and 16 represent the reducing electrophoresis results of B1, B2, B3, B4, FV1, FV2, FV3 and FV4, respectively.

FIG. 4 shows the ELISA results of bivalent bispecific antibodies with B structure and with FV structure, including the ELISA results of four B structures of B1, B2, B3 and B4 (FIG. 4A) and four FV structures of FV1, FV2, FV3 and FV4 (FIG. 4B).

FIG. 5 shows the SDS-PAGE results of the purified bispecific antibodies with preferred B2 structure and with preferred FV1 structure. M indicates Marker; Lanes 1, 2, and 3 represent the supernatants from the non-reducing transient expression of B2 bispecific antibody, the non-reducing eluent of B2 bispecific antibody by Protein L affinity chromatography, and the non-reducing eluent of B2 bispecific antibody by PD-L1 affinity chromatography; Lanes 4, 6, 8 represent the non-reducing eluent of FV1 bispecific antibody by Mab SelectSure affinity chromatography, the non-reducing eluent of FV1 bispecific antibody by Protein L affinity chromatography, and the non-reducing eluent of FV1 bispecific antibody by hCD47 affinity chromatography; and Lanes 5, 7, and 9 represent the reducing eluent of FV1 bispecific antibody by Mab SelectSure affinity chromatography, the reducing eluent of FV1 bispecific antibody by

Protein L affinity chromatography, and the reducing eluent of FV1 bispecific antibody by hCD47 affinity chromatography.

FIG. 6 shows the ELISA results of the bivalent bispecific antibodies with preferred B2 structure and with FV1 structure after purification.

DETAILED DESCRIPTION

The present disclosure discloses a bivalent bispecific antibody and a preparation method thereof, a coding gene, a host cell and a composition. In view of the content herein, those skilled in the art can make appropriate modifications to the process parameters. It should be particularly indicated that, all similar replacements and changes are obvious for those skilled in the art, which are deemed to be included in the present disclosure. The methods and uses of the present disclosure have been described by way of preferred embodiments, and it will be apparent to those skilled in the art that changes as well as appropriate modifications and combinations of the methods and uses described herein may be made without departing from the content, spirit and scope of the present disclosure, to achieve and apply the techniques of the present disclosure.

The terms used herein are explained as follows.

The term “antibody” used herein refers to an intact, monoclonal antibody. The antibody comprises two pairs of “light chain” (LC) and “heavy chain” (HC). The light chain and heavy chain of the antibody are polypeptides composed of several domains. In an intact antibody, each heavy chain comprises a heavy chain variable region VH and a heavy chain constant region. The heavy chain constant region includes the heavy chain constant domains CH1, CH2, and CH3 (antibody types IgA, IgD, and IgG) and optionally the heavy chain constant domain CH4 (antibody types IgE and IgM). Each light chain comprises a light chain variable region VL and a light chain constant region CL. An example of naturally occurring intact antibodies are the IgG antibodies. The variable regions VH and VL can be further divided into hypervariable regions, called complementarity determining regions (CDRs), and there are regions distributed between CDR which are more conservative, called framework regions (FRs). Each VH and VL consists of three CDRs and four FRs, arranged from the amino terminal to the carboxy terminal in the following order: FR1, CDR1, FR2, CDR2, FR3, CDR3, and FR4 (Janeway, C. A., Jr., et al., (2001) Immunobiology, 5th edition, Garland Press; and Woof, J., Burton D. Nat. Rev. Immunol. 4 (2004) 89-99). The two pairs of heavy and light chains can specifically bind to the same antigen. Therefore, the intact antibody is a bivalent, monospecific antibody. The “antibody” includes, for example, mouse antibodies, human antibodies, chimeric antibodies, humanized antibodies, and genetically engineered antibodies, provided that their unique properties are maintained. The particularly preferred antibodies are human or humanized antibodies, especially recombinant human or humanized antibodies.

There are five types of mammalian antibody heavy chain, represented by Greek letters: α , δ , ϵ , γ , and μ (Janeway, C. A., Jr. et al., (2001) Immunobiology, 5th edition, Garland Press). The type of heavy chain corresponds to the type of antibody, and these chains are present in IgA, IgD, IgE, IgG, and IgM antibodies, respectively (Rhoades, R. A., Pflanzner, R.G. (2002), Human Physiology, 4th edition, Thommesen knowledge). Different heavy chains differ in size and composition. The heavy chains α and γ contain about 450 amino acids, while μ and δ contain about 550 amino acids.

Each heavy chain has two types of region, the constant region and the variable region. The constant region is the same in all antibodies of the same isotype, but different in antibodies of different isotypes. The heavy chains γ , α and δ have a constant region composed of three constant domains CH1, CH2 and CH3 and a hinge region for increasing flexibility (Woof, J., Burton D. Nat. Rev. Immunol. 4 (2004) 89-99). The heavy chains μ and ϵ have a constant region composed of 4 constant domains CH1, CH2, CH3 and CH4 (Janeway, C. A., Jr. et al., (2001) Immunobiology, 5th edition, published by Garland Society). The variable region of the heavy chain differs in antibodies produced by different B cells, but is the same for all antibodies produced by a single B cell or B cell clone. The variable region of each heavy chain is approximately 110 amino acids in length and consists of a single antibody domain.

In mammals, there are only two types of light chains, which are called λ and κ . The light chain comprises two continuous domains: one constant domain CL and one variable domain VL. The light chain is approximate 211-217 amino acids in length. Preferably, the light chain is a κ light chain, and the constant domain CL is preferably CK.

The term "monoclonal antibody" or "monoclonal antibody composition" used herein refers to an antibody preparation of a single amino acid composition.

The "antibody" according to the present disclosure can be of any type (such as IgA, IgD, IgE, IgG and IgM, preferably IgG or IgE), or subtype (such as IgG1, IgG2, IgG3, IgG4, IgA1 and IgA2, preferably IgG1). The bivalent bispecific antibody according to the present disclosure is derived from two antibodies with the Fc fragments which are of the same subtype (such as IgG1, IgG4, etc., preferably IgG1), preferably Fc fragments of the same allotype.

The "Fc fragment of an antibody" or "Fc fragment" is a term well known to those skilled in the art and is defined based on the papain cleavage of the antibody. The antibody according to the present disclosure comprises, for example, Fc fragment, preferably the Fc fragment derived from a human antibody and preferably all other parts of a human constant region. The Fc fragment of the antibody is directly involved in complement activation, C1q binding, C3 activation and Fc receptor binding. Although the effect of the antibody on the complement system depends on certain conditions, the binding to C1q is caused by the binding sites in the Fc fragment. The binding sites are prior art and described in, for example, Lukas, T. J., et al., J. Immunol. 127 (1981) 2555-2560; Brunhouse, Cebra, J. J., Mo 1. Immunol. 16 (1979) 907-917; Burton, D. R., et al., Nature 288 (1980) 338-344; Thommesen, J. E., et al., Mol. Immunol. 37 (2000) 995-1004; Idusogie, E. E., et al., J. Immunol. 164 (2000) 4178-4184; Hezareh, M., et al., J. Virol. 75 (2001) 12161-12168; Morgan, A., et al., Immunology 86 (1995) 319-324; and EP0307434. The binding sites are, for example, L234, L235, D270, N297, E318, K320, K322, P331 and P329 (according to Kabat's EU catalog number). Antibodies of subtypes IgG1, IgG2, and IgG3 usually exhibit complement activation, C1q binding and C3 activation, while IgG4 neither activates the complement system, nor binds to C1q or activates C3. Preferably, the Fc fragment is a human Fc fragment.

As used herein, the term "recombinant human antibody" includes all human antibodies prepared, expressed, produced or isolated by recombinant methods, such as antibodies isolated from host cells, such as from NSO or CHO cells, or isolated from human immunoglobulin genes of transgenic animal, or antibodies expressed by recombinant expression

vectors transfected into host cells. This recombinant human antibody has constant and variable regions in a rearranged form.

As used herein, a "variable domain" refers to each pair of light chain and heavy chain that directly participates in the binding of an antibody to an antigen. The domains of human variable light chain and heavy chain have the same general structure and each domain comprises four framework regions (FR), which have generally conserved sequence and connected by three hypervariable regions (CDRs). The framework region adopts a self-folding conformation and the CDR can form a loop connecting the self-folding structure. The CDRs in each chain are maintained in their three-dimensional structure by the framework regions and form an antigen binding site together with the CDRs from the other chain. The CDR3 regions of heavy chain and light chain of antibody play a particularly important role in the binding specificity/affinity of the antibodies of the present disclosure.

As used herein, the term "hypervariable region" or "antigen-binding fragment of an antibody" refers to amino acid residues of the antibody that are responsible for antigen binding. The hypervariable region comprises amino acid residues from the "complementarity determining region". "Framework regions" are those regions of variable domains other than the hypervariable region residues defined herein. Therefore, the light chain and heavy chain of the antibody from N-terminal to C-terminal include the domains FR1, CDR1, FR2, CDR2, FR3, CDR3, and FR4, and the CDRs on each chain are separated by the framework amino acids. In particular, CDR3 of the heavy chain is the most conducive region for antigen binding. The CDR and FR regions are determined according to the standard definition of Kabat et al., Sequences of Proteins of Immunological Interest, 5th edition, Public Health Service, National Institutes of Health, Bethesda, MD, (1991).

The "constant regions" of the heavy and light chains do not directly participate in the binding of antibodies to antigens but exhibit multiple effector functions.

As used herein, the term "bivalent bispecific antibody" refers to an antibody as described above, in which two single-chain variable fragments specifically bind to different antigens, that is, the first single-chain variable fragment and the light chain constant region specifically bind to the first antigen, while the second single-chain variable fragment and the heavy chain constant region 1 and partial hinge region specifically bind to the second antigen. The bivalent bispecific antibody can specifically bind to two different antigens but no more than two antigens at the same time. In contrast, on one hand, a monospecific antibody that can only bind to one antigen, and on the other hand, a tetravalent, tetraspecific antibody that can bind four antigen at the same time.

As used herein, the terms "antigen" and "antigen molecule" are used interchangeably and refer to all molecules that can be specifically bound by an antibody. The bivalent bispecific antibody specifically binds to a first antigen and a second different antigen. As used herein, the term "antigen" includes, for example, proteins, different epitopes on the protein (as different antigens within the meaning of the present disclosure), and polysaccharides, which mainly includes bacteria, viruses and parts of other microorganisms (shell, envelope, cell wall, flagella, fimbriae and toxins). Lipids and nucleic acids are antigenic only when combined with proteins and polysaccharides. Non-microbial exogenous (non-self) antigens may include pollen, egg white, and proteins from transplanted tissues and organs or proteins on the surface of infused blood cells. Preferably, the antigen is

selected from the group consisting of cytokines, cell surface proteins, enzymes and receptor.

Tumor antigens are those antigens contained in MHC I or MHC II molecules on the surface of tumor cells. These antigens can sometimes be presented by tumor cells, and never by normal cells. As such, they are called tumor-specific antigens (TSAs) and are typically produced by tumor-specific mutations. More common are antigens presented by tumor cells and normal cells, and they are called tumor-associated antigens (TAAs). Cytotoxic T lymphocytes that recognize these antigens may be able to destroy tumor cells before they proliferate or metastasize. Tumor antigens can also be present on the tumor surface in the form of, for example, mutant receptors, in which case they should be recognized by B cells.

In a preferred embodiment, at least one of the two different antigens (the first and second antigens) to which the bivalent bispecific antibody specifically binds is a tumor antigen.

In another preferred embodiment, both of the two different antigens (the first and second antigens) to which the bivalent bispecific antibody specifically binds are tumor antigens. In this case, the first and second antigens may also be the two different epitopes on the same tumor-specific protein.

In another preferred embodiment, one of the two different antigens (the first and second antigens) to which the bivalent bispecific antibody specifically binds is a tumor antigen, and the other is an effector cell antigen, such as a T cell receptor, CD3, CD16, etc.

In another preferred embodiment, one of the two different antigens (the first and second antigens) to which the bivalent bispecific antibody specifically binds is a tumor antigen, and the other is an anti-cancer substance such as a toxin or a kinase inhibitor.

As used herein, "specific binding" or "specifically bind" refers to an antibody that specifically binds to an antigen. Preferably, the binding affinity of the antibody that specifically binds to the antigen has a KD value below 10⁻⁹ mol/L, such as 10⁻¹⁰ mol/L, preferably a KD value below 10⁻¹⁰ mol/L, such as 10⁻¹² mol/L. Binding affinity is determined using standard binding assays, such as surface plasmon resonance technology (Biacore).

The term "epitope" comprises any polypeptide determinant capable of specifically binding an antibody. In certain embodiments, epitope determinants include chemically polar surface groupings of molecules, such as amino acids, sugar side chains, phosphoryl or sulfonyl groups. In certain embodiments, an epitope can have specific three-dimensional structural characteristics and/or specific charging characteristics. An epitope is a region of an antigen bound by an antibody. In certain embodiments, when an antibody preferably recognizes its target antigen in a complex mixture of proteins and/or macromolecules, the antibody is said to specifically bind to the antigen.

As used herein, the term "nucleic acid" or "nucleic acid molecule" includes DNA molecules and RNA molecules. The nucleic acid molecule may be single-stranded or double-stranded, but is preferably double-stranded DNA.

As used herein, the expressions "cell", "cell line" and "cell culture" are used interchangeably, and all of these include progeny. Therefore, the terms "transformant" and "transformed cells" include primary subject cells and cultures derived therefrom, regardless of the number of generation. It is also understood that the DNA content of all offspring may not be exactly the same due to intentional or unintentional mutations. They also include the mutant off-

spring with the same function or biological activity screened in the originally transformed cells. When different names are meant, it will be clear due to the context.

As used herein, the term "transformation" refers to the process of transferring a vector/nucleic acid into a host cell. If cells without an insurmountable cell wall barrier are used as host cells, transfection is performed, for example, by the calcium phosphate precipitation method as described in Graham and van der Eh, *Virology* 52 (1978) 546ff. However, other methods of introducing DNA into cells can also be used, such as by nuclear injection or by protoplast fusion. If prokaryotic cells or cells containing substantial cell wall structures are used, transfection is performed, for example, by calcium treatment with calcium oxide, as described in Cohen, F. N., et al., *PNAS*. 69 (1972) 7110ff.

The use of transformation recombination to generate antibodies is well known in the prior art and is described in, for example, the review article Makrides, S. C., *Protein Expr. Purif.* 17 (1999) 183-202; Geisse, S., et al., *Protein Expr. Purif.* 8 (1996) 271-282; Kaufman, R. J., *Mol. Biotechnol.* 16 (2000) 151-161; Werner, R. G., et al., *Arzneimittel Forschung* 48 (1998) 870-880 and U.S. Pat. Nos. 6,331,415 and 4,816,567.

As used herein, "expression" refers to the process of transcribing nucleic acid into mRNA and/or the process of subsequently translating the transcribed mRNA into peptides, polypeptides or proteins. The transcript and the encoded polypeptide are collectively called the gene product. If the polynucleotide is derived from genomic DNA, expression in eukaryotic cells may include splicing of mRNA.

A "vector" is a nucleic acid molecule, especially self-replicating one, which transfers the inserted nucleic acid molecule into and/or between host cells. The term includes vectors whose primary function is to insert DNA or RNA into cells, replication vectors whose primary function is to replicate DNA or RNA, and expression vectors whose function is to transcribe and/or translate DNA or RNA. It also includes vectors that provide more than one of the above-mentioned functions.

An "expression vector" is a polynucleotide, which can be transcribed and translated into a polypeptide after being introduced into a suitable host cell. "Expression system" generally refers to a suitable host cell including an expression vector whose function is to produce a desired expression product.

The bivalent bispecific antibody according to the present disclosure is preferably produced by a recombinant method. The method is generally known in the art and includes protein expression in prokaryotic and eukaryotic cells, subsequent isolation of antibody polypeptides and generally purification to pharmaceutical purity. For protein expression, nucleic acids encoding light and heavy chains or fragments thereof are inserted in an expression vector by standard methods. The expression is carried out in suitable prokaryotic or eukaryotic host cells such as CHO cells, NSO cells, SP2/0 cells, HEK293 cells, COS cells, yeast or *E. coli* cells, and the antibody is recovered from the cells (the supernatant or cells after lysis). The bivalent bispecific antibody may exist in whole cells, as cell lysates, or in partially purified or substantially pure form. Purification is carried out by standard techniques, including alkali/SDS treatment, column chromatography and other techniques well known in the art to eliminate other cellular components or other contaminants, such as other cellular nucleic acids or proteins. Reference could be made to Ausubel, F., et al., *Current Protocols in Molecular Biology*, Greene Publishing

and WileyInter science, New York (1987). The expression in NSO cells is described in, for example, Barnes, L. M., et al., Cytotechnology 32 (2000) 109-123; and Barnes, L. M., et al., Biotech. Bioeng. 73 (2001) 261-270. Transient expression is described in, for example, Durocher, Y., et al., Nucl. Acids Res. 30 (2002) IV. The cloning of variable domains is described in Orlando, R., et al., Proc. Natl. Acad. Sci. USA 86 (1989) 3833-3837; Carter, P., et al., Proc. Natl. Acad. Sci. USA 89 (1992) 4285-4289; and Norderha, L., et al., J. Immunol. Methods 204 (1997) 77-87. A preferred transient expression system (HEK 293) is described in Schlaeger, E. J., and Christensen, K., Cytotechnology 30 (1999) 71-83 and Schlaeger, E. J., J. Immunol. Methods 194 (1996) 191-199.

Control sequences suitable for prokaryotes comprise, for example, promoters, optional operon sequences, and ribosome binding sites. It is known that eukaryotic cells utilize promoters, enhancers and polyadenylation signals.

A nucleic acid is "operably linked" when placed in a functional relationship with another nucleic acid sequence. For example, the DNA of the polypeptide is operably linked to a leader sequence or a DNA secreting leader sequence, provided that it is expressed as a preprotein involved in the secretion of the polypeptide; the coding sequence is operably linked to a promoter or enhancer, provided that it affects the transcription of the sequence; or the coding sequence is operably linked to a ribosome binding site, provided that it is positioned to facilitate translation. Generally, "operably linked" means that the DNA sequence being linked is continuous, and in the case of being linked to sequence secreting a leader, is continuous and in an open reading frame. However, the enhancer does not have to be continuous. Linkage is achieved by linking at convenient restriction sites. If the site is not present, the synthetic oligonucleotide adaptor or linker is used according to conventional practice.

By conventional immunoglobulin purification procedures, such as, for example, protein A-agarose, hydroxyapatite chromatography, gel electrophoresis, dialysis, or affinity chromatography, the bivalent bispecific antibody will be appropriately isolated from the culture medium. The DNA and RNA encoding monoclonal antibodies can be easily isolated and sequenced using conventional procedures. Quadroma cells can function as the source of the DNA and RNA. Once isolated, the DNA can be inserted into an expression vector, which is then transfected into host cells that would otherwise not produce immunoglobulin, such as HEK293 cells, CHO cells, or myeloma cells, to generate the recombinant monoclonal antibodies in the host cells.

Amino acid sequence variants (or mutants) of bivalent bispecific antibodies are prepared by introducing appropriate nucleotide changes into the antibody DNA, or by nucleotide synthesis. However, such modification can only be performed in a very limited range, such as the range described above. In addition, the modification does not change the above-mentioned antibody characteristics, such as IgG isotype and antigen binding, but can improve the yield of recombinant products or protein stability or promote purification.

Raw materials or reagents, used in the bivalent bispecific antibody and the preparation method thereof, the coding gene, host cell and composition provided by the present disclosure, can all be purchased from the market.

The present disclosure will be further illustrated by the following examples:

Example 1 Preparation of Bivalent Bispecific Antibodies

1. Construction of transient transfection expression vectors for bivalent bispecific antibodies

(1) Materials

The sequence of VL (SEQ ID NO: 18) and the sequence of VH (SEQ ID NO: 19) are derived from the anti-human CD47 humanized monoclonal antibody 059-4.16.2 H1L2, which was obtained by humanization of murine antibody (CN 201610436519.3) obtained from quadromas.

The sequence of VL (SEQ ID NO: 20) and the sequence of VH (SEQ ID NO: 21) was derived from the anti-human PD-L1 humanized monoclonal antibody 047 Ab-6, which was obtained by panning the natural human antibodies library (CN 201810044303.1). The coding nucleotides of heavy chain constant region CH1, hinge region and Fc of IgG1, and nucleotides of Kappa chain constant region are derived from human IgG1.

(2) Methods

pGS003 was selected to construct the expression vectors for the heavy chain and light chain of bivalent bispecific antibodies (8 antibodies, the schematic diagrams of the structure are shown in FIG. 1 and FIG. 2). Primers were designed according to the coding nucleotides of VL and VH derived from the anti-human CD47 humanized monoclonal antibody 059-4.16.2 H1L2, the coding nucleotides of VL and VH derived from the anti-human PD-L1 humanized monoclonal antibody 047 Ab-6, the coding nucleotides of heavy chain constant region CH1, hinge region and Fc of IgG1, and nucleotide sequence of Kappa chain constant region, and multiple cloning sites in the vector. After PCR amplification, 9 heavy chain coding sequences and 8 light chain coding sequences were respectively cloned into pGS003 by in vitro recombination method (Synbio Technologies, iMulli Multi-fragment Recombination Cloning Kit), as shown in Table 1. After sequencing to identify the correct insertion of the antibody gene, the recombinant expression vector was transformed into *E. coli* TOP10F'. Then single colonies was picked and inoculated in LB medium containing 100 µg/mL of ampicillin, and cultured with shaking at 37° C. for 16 hours. The plasmids were extracted using endotoxin-removal, large-scale extraction kit of Zymo Research, and the obtained plasmids were dissolved in 1 mL of ultrapure water, and the plasmid concentration and OD260/280 were determined with a spectrophotometer. A plasmid with OD260/280 value between 1.8 and 1.9 is relatively pure plasmid DNA.

TABLE 1

List of transient transfection expression vectors for heavy and light chains			
Heavy chain expression vector	Heavy chain amino acid sequence	Light chain expression vector	Light chain amino acid sequence
H1	SEQ ID NO: 1	L1	SEQ ID NO: 10
H2	SEQ ID NO: 2	L2	SEQ ID NO: 11
H3	SEQ ID NO: 3	L3	SEQ ID NO: 12
H4	SEQ ID NO: 4	L4	SEQ ID NO: 13
H5	SEQ ID NO: 5	L5	SEQ ID NO: 14
H6	SEQ ID NO: 6	L6	SEQ ID NO: 15
H7	SEQ ID NO: 7	L7	SEQ ID NO: 16
H8	SEQ ID NO: 8	L8	SEQ ID NO: 17
H9	SEQ ID NO: 9	/	/

2. Transfection, Expression and Detection in Mammalian 293E Cells

In the above 9 heavy chain expression vectors and 8 light chain expression vectors, H1, L2, L3 and H4 were used to express scFv of anti-hCD47, L1, H2, H3 and L4 were used to express scFv of anti-hPD-L1, H5 and L5 were used to express VH and VL of anti-hPD-L1, respectively, H6, H7,

15

H8 and H9 were used to express VH of anti-hCD47, and L6, L7 and L8 were used to express VL of anti-hCD47.

After combining the above vectors in accordance with H1+L1 (B1 structure), H2+L2 (B2 structure), H3+L3 (B3 structure), H4+L4 (B4 structure), H5+L5+H6+L6 (FV1 structure), H5+L5+H7+L7 (FV2 structure), H5+L5+H8+L8 (FV3 structure) and H5+L5+H9+L7 (FV4 structure), 2 mL 293E system of transient transfection expression was performed for evaluation. As to flexible peptide and linker, FV1 was L/GGGC (LGGC (SEQ ID NO: 27) or GGGC (SEQ ID NO: 28)), FV2 was GGGGS (SEQ ID NO: 25) or GGGG-SAS (SEQ ID NO: 26)+LGGC (SEQ ID NO: 27) or GGGC (SEQ ID NO: 28), FV3 was GGGGSGGGGSGGGGS (SEQ ID NO: 29) or GGGGSASGGGGSGGGGSAS (SEQ ID NO: 30)+LGGC (SEQ ID NO: 27) or GGGC (SEQ ID NO: 28), and for FV4, the light chain was GGGGS (SEQ ID NO: 25) or GGGGSAS (SEQ ID NO: 26)+LGGC (SEQ ID NO: 27) or GGGC (SEQ ID NO: 28) and the heavy chain was LGGC (SEQ ID NO: 27) or GGGC (SEQ ID NO: 28)+GGGGS (SEQ ID NO: 25) or GGGGSAS (SEQ ID NO: 26). The expression level was detected and the binding between the antibody and human CD47 or human PD-L1 was detected by ELISA. The results are shown in FIG. 3 and FIG. 4 (wherein 4A represents the ELISA detection result of the bivalent bispecific antibody with B structure, and 4B represents the ELISA detection result of the bivalent bispecific antibody with FV structure). B2 structure performs the best in the expression, assembly and antigen binding among the bispecific antibodies with B-type structure; and FV1 structure performs the best in the expression, assembly and antigen binding among the bispecific antibodies with FV-type structure. Therefore, B2 and FV1 are preferred antibodies.

293E cells were used to perform amplified transient transfection expression of B2 and FV1 structures in Free-style medium. 24 hours before transfection, 300 mL of 293E cells at 0.5×10^6 cells/mL were seeded in a 1 L cell culture flask, and cultured in a 37° C., 5% CO₂ incubator with shaking at 120 rpm. During transfection, 300 µL of 293fectin™ was added to 5.7 mL Opti-MEM™. After mixing well, the mixture was incubated at room temperature for 2 minutes. Meanwhile, 300 µg of the expression plasmids for B2 and FV1 were diluted to 6 mL with Opti-MEM™, respectively. The diluted transfection reagent 293fectin and plasmids were mixed thoroughly and incubated at room temperature for 15 minutes. After that, the mixture was added to cells and mixed well, and cultured in a 37° C., 5% CO₂ incubator with shaking at 120 rpm for 7 days.

Example 2 Purification and Detection of Preferred Antibodies

Purification of Proteins with B2 Structure

The cell culture medium was centrifuged at 2000 g for 20 min, and the supernatant was collected and then filtered with a 0.22 micron filter membrane. Next, the supernatant was subjected to Protein L (GE) chromatography, the proteins were eluted with 20 mM citrate-sodium citrate, pH 3.0, and then the resultant was adjusted to neutral pH with 1 M Tris base. After protein L chromatography, the sample was then subjected to affinity chromatography coupled with human PDL1 protein, the proteins were eluted with 20 mM citrate-sodium citrate, pH 3.0, and then the resultant was adjusted to neutral pH with 1 M Tris base. Purified samples were separated by SDS-PAGE using 4-20% gradient gel (GenScript Biotechnology Co., Ltd.) to detect purified proteins.

16

The results are shown in FIG. 5 and the purity of the preferred antibody B2 was 95%.

Purification of Proteins with FV1 Structure

The cell culture medium was centrifuged at 2000 g for 20 min, and the supernatant was collected and then filtered with a 0.22 micron filter membrane. Next, the supernatant was subjected to Mabselect Sure (GE) chromatography, the proteins were eluted with 20 mM citrate-sodium citrate, pH 3.0, and then the resultant was adjusted to neutral pH with 1 M Tris base. After Mabselect Sure chromatography, the sample was then subjected to Protein L (GE) chromatography, the proteins were eluted with 20 mM citrate-sodium citrate, pH 3.0, and then the resultant was adjusted to neutral pH with 1 M Tris base. After protein L chromatography, the sample was then subjected to affinity chromatography coupled with human CD47 protein, the proteins were eluted with 20 mM citrate-sodium citrate, pH 3.0, and then the resultant was adjusted to neutral pH with 1 M Tris base. Purified samples were separated by SDS-PAGE using 4-20% gradient gel (GenScript Biotechnology Co., Ltd.) to detect purified proteins. The results are shown in FIG. 5 and the purity of the preferred FV1 was 90.8%.

Example 3 ELISA Detection of Preferred Antibodies Binding to Human CD47 and Human PD-L1

1. Coating the first antigen: Human PD-L1-mFc (constructed by GeneScience, SEQ ID NO: 22) was diluted with PBS to 1 µg/mL, and then added to a 96-well microtiter plate at 50 µL per well and incubated overnight at 4° C.
2. Blocking: After being washed three times, the plate was blocked with 3% BSA at 250 µL per well, and incubated at 37° C. for 2 hours.
3. Adding candidate antibody: After washing three times, the candidate antibody was added to the plate, each with 12 samples diluted at a 2-fold concentration gradient with an initial concentration of 10 mg/mL, positive control or negative control was added at 50 µL per well, and incubated at 25° C. for 1 hour.
4. Adding the second antigen: Human CD47-His (constructed by GeneScience, SEQ ID NO: 23) was diluted with PBS to 10 µg/mL, and then added to the 96-well microtiter plate at 50 µL per well and incubated at 25° C. for 1 hour.
5. Adding the secondary antibody: After washing three times, HRP-labeled streptavidin (1:10,000) was added to the plate at 50 µL per well, and incubated at 25° C. for 1 hour.
6. Color development: After being washed four times, TMB color development solution was added to the plate 50 µL per well, and developed color shielded from light at room temperature for 10 minutes.
7. Terminating: The stop solution was directly added to the plate at 50 µL per well to terminate the reaction.
8. Detection: After terminating the reaction, the microtiter plate was immediately put into the microplate reader. The OD value at 450 nm was measured, and the original data was saved for sorting. The results are shown in FIG. 6, showing that for the purified antibody B2, EC50=0.5636; for FV1, EC50=1.662.

Example 4 Affinity Determination of Preferred Antibodies

The affinities of antibodies with B2 structure and with FV1 structure were detected by Biacore T200 instrument. The specific protocols were as follows. Human PD-L1-His and human CD47-His were coupled to CM5 biosensor chip

17

(GE), and then the antibodies of different concentrations were flowed through the chip at a flow rate of 30 $\mu\text{L}/\text{min}$. The binding between the candidate antibody and antigen was performed with a binding time of 120 s and a dissociation time of 300 s. The kinetic fitting was performed using BIAevaluation software (GE), and the results of affinity constants were obtained as shown in Table 2 and Table 3. The affinities of B2 and FV1 with PD-L1 were 1.09E-10 M and 2.71E-10 M, respectively; and the affinities of B2 and FV1 with CD47 were 2.58E-8 M and 1.50E-8 M, respectively.

TABLE 2

Results of affinity determination of candidate antibodies with PD-L1				
Antibody	Ka (1/Ms)	Kd (1/s)	KD (M)	Rmax (RU)
B2	3.95E+05	4.28E-05	1.09E-10	7.067
FV1	4.40E+05	1.19E-04	2.71E-10	28.06
PD-L1 positive monoclonal antibody	9.08E+05	2.36E-04	2.60E-10	80

18

TABLE 3

Results of affinity determination of candidate antibodies with CD47		
Antibody	KD (M)	Rmax (RU)
B2	2.58E-08	50.00
FV1	1.50E-08	60.00
CD47 positive monoclonal antibody	2.36E-08	50.92

The foregoing is preferred embodiments of the present disclosure, however, it should be noted that some improvements and modifications can be made thereto by those ordinary skilled in the art without departing from the principles of the present disclosure, and these improvements and modifications should also be deemed to be within the protection scope of the present disclosure.

SEQUENCE LISTING

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<160> NUMBER OF SEQ ID NOS: 30

<210> SEQ ID NO 1
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<212> TYPE: PRT
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Heavy chain amino acid sequence

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Ala Ser Thr Lys Gly Pro Ser Val Phe Pro Leu Ala Pro Ser Ser Lys
1          5          10          15

Ser Thr Ser Gly Gly Thr Ala Ala Leu Gly Cys Leu Val Lys Asp Tyr
20          25          30

Phe Pro Glu Pro Val Thr Val Ser Trp Asn Ser Gly Ala Leu Thr Ser
35          40          45

Gly Val His Thr Phe Pro Ala Val Leu Gln Ser Ser Gly Leu Tyr Ser
50          55          60

Leu Ser Ser Val Val Thr Val Pro Ser Ser Ser Leu Gly Thr Gln Thr
65          70          75          80

Tyr Ile Cys Asn Val Asn His Lys Pro Ser Asn Thr Lys Val Asp Lys
85          90          95

Lys Val Glu Pro Lys Ser Cys Asp Lys Gln Val Gln Leu Val Gln Ser
100         105         110

Gly Ala Glu Val Lys Lys Pro Gly Ala Ser Val Lys Val Ser Cys Lys
115         120         125

Ala Ser Gly Tyr Thr Phe Thr Asn Tyr Val Leu His Trp Val Arg Gln
130         135         140

Ala Pro Gly Gln Arg Leu Glu Trp Met Gly Tyr Phe Asn Pro Tyr Asn
145         150         155         160

Asp Asp Ser Lys Tyr Asn Glu Lys Phe Lys Gly Arg Val Thr Ile Thr
165         170         175

Arg Asp Thr Ser Ala Ser Thr Ala Tyr Met Glu Leu Ser Ser Leu Arg
180         185         190

Ser Glu Asp Thr Ala Val Tyr Tyr Cys Ala Arg Gly Gly Tyr Tyr Ser
195         200         205

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Met Asp Tyr Trp Gly Gln Gly Thr Leu Val Thr Val Ser Ser Gly Gly
 210                215                220

Gly Gly Ser Gly Gly Gly Gly Ser Gly Gly Gly Gly Ser Ala Ser Asp
225                230                235                240

Ile Val Met Thr Gln Thr Pro Leu Ser Leu Pro Val Thr Pro Gly Gln
                245                250                255

Pro Ala Ser Ile Ser Cys Arg Ala Ser Gln Ser Leu Ala His Ser Asn
                260                265                270

Gly Tyr Thr Tyr Leu Gln Trp Tyr Leu Gln Lys Pro Gly Gln Ser Pro
                275                280                285

Gln Leu Leu Ile Tyr Lys Val Ser Lys Arg Phe Ser Gly Val Pro Asp
                290                295                300

Arg Phe Ser Gly Ser Gly Ser Gly Thr Asp Phe Thr Leu Lys Ile Ser
305                310                315                320

Arg Val Glu Ala Glu Asp Val Gly Val Tyr Tyr Cys Ser Gln Ser Thr
                325                330                335

His Val Pro Tyr Thr Phe Gly Gln Gly Thr Lys Val Glu Ile Lys
                340                345                350

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<210> SEQ ID NO 2
<211> LENGTH: 347
<212> TYPE: PRT
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Heavy chain amino acid sequence

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<400> SEQUENCE: 2

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Ala Ser Thr Lys Gly Pro Ser Val Phe Pro Leu Ala Pro Ser Ser Lys
 1          5          10          15

Ser Thr Ser Gly Gly Thr Ala Ala Leu Gly Cys Leu Val Lys Asp Tyr
 20          25          30

Phe Pro Glu Pro Val Thr Val Ser Trp Asn Ser Gly Ala Leu Thr Ser
 35          40          45

Gly Val His Thr Phe Pro Ala Val Leu Gln Ser Ser Gly Leu Tyr Ser
 50          55          60

Leu Ser Ser Val Val Thr Val Pro Ser Ser Ser Leu Gly Thr Gln Thr
 65          70          75          80

Tyr Ile Cys Asn Val Asn His Lys Pro Ser Asn Thr Lys Val Asp Lys
 85          90          95

Lys Val Glu Pro Lys Ser Cys Asp Lys Glu Val Gln Leu Val Glu Ser
100         105         110

Gly Gly Gly Leu Val Gln Pro Gly Gly Ser Leu Arg Leu Ser Cys Ala
115         120         125

Ala Ser Tyr Phe Pro Phe Ser Asp Ser Trp Ile His Trp Val Arg Gln
130         135         140

Ala Pro Gly Lys Gly Leu Glu Trp Val Ala Trp Ile Ser Pro Tyr Gly
145         150         155         160

Gly Ser Ser Tyr Tyr Ala Asp Ser Val Gln Asp Arg Phe Thr Ile Ser
165         170         175

Ala Asp Thr Ser Lys Asn Thr Ala Tyr Leu Gln Met Asn Ser Leu Arg
180         185         190

Ala Glu Asp Thr Ala Val Tyr Tyr Cys Ala Lys Arg His Trp Pro Gly
195         200         205

Gly Phe Asp His Trp Gly Gln Gly Thr Leu Val Thr Val Ser Ser Gly
210         215         220

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Gly Gly Gly Ser Gly Gly Gly Gly Ser Gly Gly Gly Gly Ser Ala Ser
 225 230 235 240
 Asp Ile Gln Met Thr Gln Ser Pro Ser Ser Leu Ser Ala Ser Val Gly
 245 250 255
 Asp Arg Val Thr Ile Thr Cys Ser Ala Ser Gln Asp Val Thr Thr Ala
 260 265 270
 Val Ala Trp Tyr Gln Gln Lys Pro Gly Lys Ala Pro Lys Leu Leu Ile
 275 280 285
 Tyr Ser Ala Ser Phe Pro Tyr Arg Gly Val Pro Ser Arg Phe Ser Gly
 290 295 300
 Ser Gly Ser Gly Thr Asp Phe Thr Leu Thr Ile Ser Ser Leu Gln Pro
 305 310 315 320
 Glu Asp Phe Ala Thr Tyr Tyr Cys Gln Gln Tyr Met Tyr His Pro Ser
 325 330 335
 Thr Phe Gly Gln Gly Thr Lys Val Glu Ile Lys
 340 345

<210> SEQ ID NO 3
 <211> LENGTH: 347
 <212> TYPE: PRT
 <213> ORGANISM: Artificial sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Heavy chain amino acid sequence

<400> SEQUENCE: 3

Glu Val Gln Leu Val Glu Ser Gly Gly Gly Leu Val Gln Pro Gly Gly
 1 5 10 15
 Ser Leu Arg Leu Ser Cys Ala Ala Ser Tyr Phe Pro Phe Ser Asp Ser
 20 25 30
 Trp Ile His Trp Val Arg Gln Ala Pro Gly Lys Gly Leu Glu Trp Val
 35 40 45
 Ala Trp Ile Ser Pro Tyr Gly Gly Ser Ser Tyr Tyr Ala Asp Ser Val
 50 55 60
 Gln Asp Arg Phe Thr Ile Ser Ala Asp Thr Ser Lys Asn Thr Ala Tyr
 65 70 75 80
 Leu Gln Met Asn Ser Leu Arg Ala Glu Asp Thr Ala Val Tyr Tyr Cys
 85 90 95
 Ala Lys Arg His Trp Pro Gly Gly Phe Asp His Trp Gly Gln Gly Thr
 100 105 110
 Leu Val Thr Val Ser Ser Gly Gly Gly Gly Ser Gly Gly Gly Gly Ser
 115 120 125
 Gly Gly Gly Gly Ser Ala Ser Asp Ile Gln Met Thr Gln Ser Pro Ser
 130 135 140
 Ser Leu Ser Ala Ser Val Gly Asp Arg Val Thr Ile Thr Cys Ser Ala
 145 150 155 160
 Ser Gln Asp Val Thr Thr Ala Val Ala Trp Tyr Gln Gln Lys Pro Gly
 165 170 175
 Lys Ala Pro Lys Leu Leu Ile Tyr Ser Ala Ser Phe Pro Tyr Arg Gly
 180 185 190
 Val Pro Ser Arg Phe Ser Gly Ser Gly Ser Gly Thr Asp Phe Thr Leu
 195 200 205
 Thr Ile Ser Ser Leu Gln Pro Glu Asp Phe Ala Thr Tyr Tyr Cys Gln
 210 215 220
 Gln Tyr Met Tyr His Pro Ser Thr Phe Gly Gln Gly Thr Lys Val Glu
 225 230 235 240

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Ile Lys Ala Ser Thr Lys Gly Pro Ser Val Phe Pro Leu Ala Pro Ser
      245                      250                      255
Ser Lys Ser Thr Ser Gly Gly Thr Ala Ala Leu Gly Cys Leu Val Lys
      260                      265                      270
Asp Tyr Phe Pro Glu Pro Val Thr Val Ser Trp Asn Ser Gly Ala Leu
      275                      280                      285
Thr Ser Gly Val His Thr Phe Pro Ala Val Leu Gln Ser Ser Gly Leu
      290                      295                      300
Tyr Ser Leu Ser Ser Val Val Thr Val Pro Ser Ser Ser Leu Gly Thr
      305                      310                      315                      320
Gln Thr Tyr Ile Cys Asn Val Asn His Lys Pro Ser Asn Thr Lys Val
      325                      330                      335
Asp Lys Lys Val Glu Pro Lys Ser Cys Asp Lys
      340                      345

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<210> SEQ ID NO 4
<211> LENGTH: 351
<212> TYPE: PRT
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Heavy chain amino acid sequence

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<400> SEQUENCE: 4

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Gln Val Gln Leu Val Gln Ser Gly Ala Glu Val Lys Lys Pro Gly Ala
1      5      10      15
Ser Val Lys Val Ser Cys Lys Ala Ser Gly Tyr Thr Phe Thr Asn Tyr
      20      25      30
Val Leu His Trp Val Arg Gln Ala Pro Gly Gln Arg Leu Glu Trp Met
      35      40      45
Gly Tyr Phe Asn Pro Tyr Asn Asp Asp Ser Lys Tyr Asn Glu Lys Phe
      50      55      60
Lys Gly Arg Val Thr Ile Thr Arg Asp Thr Ser Ala Ser Thr Ala Tyr
      65      70      75      80
Met Glu Leu Ser Ser Leu Arg Ser Glu Asp Thr Ala Val Tyr Tyr Cys
      85      90      95
Ala Arg Gly Gly Tyr Tyr Ser Met Asp Tyr Trp Gly Gln Gly Thr Leu
      100     105     110
Val Thr Val Ser Ser Gly Gly Gly Gly Ser Gly Gly Gly Ser Gly
      115     120     125
Gly Gly Gly Ser Ala Ser Asp Ile Val Met Thr Gln Thr Pro Leu Ser
      130     135     140
Leu Pro Val Thr Pro Gly Gln Pro Ala Ser Ile Ser Cys Arg Ala Ser
      145     150     155     160
Gln Ser Leu Ala His Ser Asn Gly Tyr Thr Tyr Leu Gln Trp Tyr Leu
      165     170     175
Gln Lys Pro Gly Gln Ser Pro Gln Leu Leu Ile Tyr Lys Val Ser Lys
      180     185     190
Arg Phe Ser Gly Val Pro Asp Arg Phe Ser Gly Ser Gly Ser Gly Thr
      195     200     205
Asp Phe Thr Leu Lys Ile Ser Arg Val Glu Ala Glu Asp Val Gly Val
      210     215     220
Tyr Tyr Cys Ser Gln Ser Thr His Val Pro Tyr Thr Phe Gly Gln Gly
      225     230     235     240
Thr Lys Val Glu Ile Lys Ala Ser Thr Lys Gly Pro Ser Val Phe Pro
      245     250     255

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Leu Ala Pro Ser Ser Lys Ser Thr Ser Gly Gly Thr Ala Ala Leu Gly
 260 265 270
 Cys Leu Val Lys Asp Tyr Phe Pro Glu Pro Val Thr Val Ser Trp Asn
 275 280 285
 Ser Gly Ala Leu Thr Ser Gly Val His Thr Phe Pro Ala Val Leu Gln
 290 295 300
 Ser Ser Gly Leu Tyr Ser Leu Ser Ser Val Val Thr Val Pro Ser Ser
 305 310 315 320
 Ser Leu Gly Thr Gln Thr Tyr Ile Cys Asn Val Asn His Lys Pro Ser
 325 330 335
 Asn Thr Lys Val Asp Lys Lys Val Glu Pro Lys Ser Cys Asp Lys
 340 345 350

<210> SEQ ID NO 5
 <211> LENGTH: 447
 <212> TYPE: PRT
 <213> ORGANISM: Artificial sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Heavy chain amino acid sequence

<400> SEQUENCE: 5

Glu Val Gln Leu Val Glu Ser Gly Gly Gly Leu Val Gln Pro Gly Gly
 1 5 10 15
 Ser Leu Arg Leu Ser Cys Ala Ala Ser Tyr Phe Pro Phe Ser Asp Ser
 20 25 30
 Trp Ile His Trp Val Arg Gln Ala Pro Gly Lys Gly Leu Glu Trp Val
 35 40 45
 Ala Trp Ile Ser Pro Tyr Gly Gly Ser Ser Tyr Tyr Ala Asp Ser Val
 50 55 60
 Gln Asp Arg Phe Thr Ile Ser Ala Asp Thr Ser Lys Asn Thr Ala Tyr
 65 70 75 80
 Leu Gln Met Asn Ser Leu Arg Ala Glu Asp Thr Ala Val Tyr Tyr Cys
 85 90 95
 Ala Lys Arg His Trp Pro Gly Gly Phe Asp His Trp Gly Gln Gly Thr
 100 105 110
 Leu Val Thr Val Ser Ser Ala Ser Thr Lys Gly Pro Ser Val Phe Pro
 115 120 125
 Leu Ala Pro Ser Ser Lys Ser Thr Ser Gly Gly Thr Ala Ala Leu Gly
 130 135 140
 Cys Leu Val Lys Asp Tyr Phe Pro Glu Pro Val Thr Val Ser Trp Asn
 145 150 155 160
 Ser Gly Ala Leu Thr Ser Gly Val His Thr Phe Pro Ala Val Leu Gln
 165 170 175
 Ser Ser Gly Leu Tyr Ser Leu Ser Ser Val Val Thr Val Pro Ser Ser
 180 185 190
 Ser Leu Gly Thr Gln Thr Tyr Ile Cys Asn Val Asn His Lys Pro Ser
 195 200 205
 Asn Thr Lys Val Asp Lys Lys Val Glu Pro Lys Ser Cys Asp Lys Thr
 210 215 220
 His Thr Cys Pro Pro Cys Pro Ala Pro Glu Leu Leu Gly Gly Pro Ser
 225 230 235 240
 Val Phe Leu Phe Pro Pro Lys Pro Lys Asp Thr Leu Met Ile Ser Arg
 245 250 255
 Thr Pro Glu Val Thr Cys Val Val Val Asp Val Ser His Glu Asp Pro
 260 265 270

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Glu Val Lys Phe Asn Trp Tyr Val Asp Gly Val Glu Val His Asn Ala
 275 280 285
 Lys Thr Lys Pro Arg Glu Glu Gln Tyr Asn Ser Thr Tyr Arg Val Val
 290 295 300
 Ser Val Leu Thr Val Leu His Gln Asp Trp Leu Asn Gly Lys Glu Tyr
 305 310 315 320
 Lys Cys Lys Val Ser Asn Lys Ala Leu Pro Ala Pro Ile Glu Lys Thr
 325 330 335
 Ile Ser Lys Ala Lys Gly Gln Pro Arg Glu Pro Gln Val Tyr Thr Leu
 340 345 350
 Pro Pro Ser Arg Asp Glu Leu Thr Lys Asn Gln Val Ser Leu Ser Cys
 355 360 365
 Ala Val Lys Gly Phe Tyr Pro Ser Asp Ile Ala Val Glu Trp Glu Ser
 370 375 380
 Asn Gly Gln Pro Glu Asn Asn Tyr Lys Thr Thr Pro Pro Val Leu Asp
 385 390 395 400
 Ser Asp Gly Ser Phe Phe Leu Val Ser Lys Leu Thr Val Asp Lys Ser
 405 410 415
 Arg Trp Gln Gln Gly Asn Val Phe Ser Cys Ser Val Met His Glu Ala
 420 425 430
 Leu His Asn His Tyr Thr Gln Lys Ser Leu Ser Leu Ser Pro Gly
 435 440 445

<210> SEQ ID NO 6
 <211> LENGTH: 352
 <212> TYPE: PRT
 <213> ORGANISM: Artificial sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Heavy chain amino acid sequence

<400> SEQUENCE: 6

Glu Val Gln Leu Gln Gln Ser Gly Pro Glu Leu Val Lys Pro Gly Ala
 1 5 10 15
 Ser Val Arg Met Ser Cys Lys Ala Ser Gly Tyr Thr Phe Thr Asn Tyr
 20 25 30
 Val Met His Trp Val Lys Gln Lys Pro Gly Gln Gly Leu Glu Trp Ile
 35 40 45
 Gly Tyr Ile Asn Pro Tyr Asn Asp Gly Thr Lys Tyr Ser Glu Lys Phe
 50 55 60
 Lys Gly Lys Ala Thr Leu Thr Ser Asp Lys Ser Ser Ile Thr Ala Tyr
 65 70 75 80
 Met Glu Leu Ser Ser Leu Thr Ser Glu Asp Ser Ala Val Tyr Tyr Cys
 85 90 95
 Ala Lys Gly Gly Leu Phe Thr Phe Asp Tyr Trp Gly Gln Gly Thr Thr
 100 105 110
 Leu Thr Val Ser Ser Leu Gly Gly Cys Glu Pro Lys Ser Ser Asp Lys
 115 120 125
 Thr His Thr Cys Pro Pro Cys Pro Ala Pro Glu Leu Leu Gly Gly Pro
 130 135 140
 Ser Val Phe Leu Phe Pro Pro Lys Pro Lys Asp Thr Leu Met Ile Ser
 145 150 155 160
 Arg Thr Pro Glu Val Thr Cys Val Val Val Asp Val Ser His Glu Asp
 165 170 175
 Pro Glu Val Lys Phe Asn Trp Tyr Val Asp Gly Val Glu Val His Asn
 180 185 190

-continued

Ala Lys Thr Lys Pro Arg Glu Glu Gln Tyr Asn Ser Thr Tyr Arg Val
 195 200 205

Val Ser Val Leu Thr Val Leu His Gln Asp Trp Leu Asn Gly Lys Glu
 210 215 220

Tyr Lys Cys Lys Val Ser Asn Lys Ala Leu Pro Ala Pro Ile Glu Lys
 225 230 235 240

Thr Ile Ser Lys Ala Lys Gly Gln Pro Arg Glu Pro Gln Val Tyr Thr
 245 250 255

Leu Pro Pro Ser Arg Asp Glu Leu Thr Lys Asn Gln Val Ser Leu Trp
 260 265 270

Cys Leu Val Lys Gly Phe Tyr Pro Ser Asp Ile Ala Val Glu Trp Glu
 275 280 285

Ser Asn Gly Gln Pro Glu Asn Asn Tyr Lys Thr Thr Pro Pro Val Leu
 290 295 300

Asp Ser Asp Gly Ser Phe Phe Leu Tyr Ser Lys Leu Thr Val Asp Lys
 305 310 315 320

Ser Arg Trp Gln Gln Gly Asn Val Phe Ser Cys Ser Val Met His Glu
 325 330 335

Ala Leu His Asn His Tyr Thr Gln Lys Ser Leu Ser Leu Ser Pro Gly
 340 345 350

<210> SEQ ID NO 7
 <211> LENGTH: 360
 <212> TYPE: PRT
 <213> ORGANISM: Artificial sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Heavy chain amino acid sequence

<400> SEQUENCE: 7

Glu Val Gln Leu Gln Gln Ser Gly Pro Glu Leu Val Lys Pro Gly Ala
 1 5 10 15

Ser Val Arg Met Ser Cys Lys Ala Ser Gly Tyr Thr Phe Thr Asn Tyr
 20 25 30

Val Met His Trp Val Lys Gln Lys Pro Gly Gln Gly Leu Glu Trp Ile
 35 40 45

Gly Tyr Ile Asn Pro Tyr Asn Asp Gly Thr Lys Tyr Ser Glu Lys Phe
 50 55 60

Lys Gly Lys Ala Thr Leu Thr Ser Asp Lys Ser Ser Ile Thr Ala Tyr
 65 70 75 80

Met Glu Leu Ser Ser Leu Thr Ser Glu Asp Ser Ala Val Tyr Tyr Cys
 85 90 95

Ala Lys Gly Gly Leu Phe Thr Phe Asp Tyr Trp Gly Gln Gly Thr Thr
 100 105 110

Leu Thr Val Ser Ser Ser Gly Gly Gly Gly Ser Ala Ser Leu Gly Gly
 115 120 125

Cys Glu Pro Lys Ser Ser Asp Lys Thr His Thr Cys Pro Pro Cys Pro
 130 135 140

Ala Pro Glu Leu Leu Gly Gly Pro Ser Val Phe Leu Phe Pro Pro Lys
 145 150 155 160

Pro Lys Asp Thr Leu Met Ile Ser Arg Thr Pro Glu Val Thr Cys Val
 165 170 175

Val Val Asp Val Ser His Glu Asp Pro Glu Val Lys Phe Asn Trp Tyr
 180 185 190

Val Asp Gly Val Glu Val His Asn Ala Lys Thr Lys Pro Arg Glu Glu
 195 200 205

-continued

Gln Tyr Asn Ser Thr Tyr Arg Val Val Ser Val Leu Thr Val Leu His
 210 215 220

Gln Asp Trp Leu Asn Gly Lys Glu Tyr Lys Cys Lys Val Ser Asn Lys
 225 230 235 240

Ala Leu Pro Ala Pro Ile Glu Lys Thr Ile Ser Lys Ala Lys Gly Gln
 245 250 255

Pro Arg Glu Pro Gln Val Tyr Thr Leu Pro Pro Ser Arg Asp Glu Leu
 260 265 270

Thr Lys Asn Gln Val Ser Leu Trp Cys Leu Val Lys Gly Phe Tyr Pro
 275 280 285

Ser Asp Ile Ala Val Glu Trp Glu Ser Asn Gly Gln Pro Glu Asn Asn
 290 295 300

Tyr Lys Thr Thr Pro Pro Val Leu Asp Ser Asp Gly Ser Phe Phe Leu
 305 310 315 320

Tyr Ser Lys Leu Thr Val Asp Lys Ser Arg Trp Gln Gln Gly Asn Val
 325 330 335

Phe Ser Cys Ser Val Met His Glu Ala Leu His Asn His Tyr Thr Gln
 340 345 350

Lys Ser Leu Ser Leu Ser Pro Gly
 355 360

<210> SEQ ID NO 8
 <211> LENGTH: 369
 <212> TYPE: PRT
 <213> ORGANISM: Artificial sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Heavy chain amino acid sequence

<400> SEQUENCE: 8

Glu Val Gln Leu Gln Gln Ser Gly Pro Glu Leu Val Lys Pro Gly Ala
 1 5 10 15

Ser Val Arg Met Ser Cys Lys Ala Ser Gly Tyr Thr Phe Thr Asn Tyr
 20 25 30

Val Met His Trp Val Lys Gln Lys Pro Gly Gln Gly Leu Glu Trp Ile
 35 40 45

Gly Tyr Ile Asn Pro Tyr Asn Asp Gly Thr Lys Tyr Ser Glu Lys Phe
 50 55 60

Lys Gly Lys Ala Thr Leu Thr Ser Asp Lys Ser Ser Ile Thr Ala Tyr
 65 70 75 80

Met Glu Leu Ser Ser Leu Thr Ser Glu Asp Ser Ala Val Tyr Tyr Cys
 85 90 95

Ala Lys Gly Gly Leu Phe Thr Phe Asp Tyr Trp Gly Gln Gly Thr Thr
 100 105 110

Leu Thr Val Ser Ser Gly Gly Gly Gly Ser Gly Gly Gly Ser Gly
 115 120 125

Gly Gly Gly Ser Ala Ser Leu Gly Gly Cys Glu Pro Lys Ser Ser Asp
 130 135 140

Lys Thr His Thr Cys Pro Pro Cys Pro Ala Pro Glu Leu Leu Gly Gly
 145 150 155 160

Pro Ser Val Phe Leu Phe Pro Pro Lys Pro Lys Asp Thr Leu Met Ile
 165 170 175

Ser Arg Thr Pro Glu Val Thr Cys Val Val Val Asp Val Ser His Glu
 180 185 190

Asp Pro Glu Val Lys Phe Asn Trp Tyr Val Asp Gly Val Glu Val His
 195 200 205

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Asn Ala Lys Thr Lys Pro Arg Glu Glu Gln Tyr Asn Ser Thr Tyr Arg
 210                      215                      220

Val Val Ser Val Leu Thr Val Leu His Gln Asp Trp Leu Asn Gly Lys
 225                      230                      235                      240

Glu Tyr Lys Cys Lys Val Ser Asn Lys Ala Leu Pro Ala Pro Ile Glu
                      245                      250                      255

Lys Thr Ile Ser Lys Ala Lys Gly Gln Pro Arg Glu Pro Gln Val Tyr
                      260                      265                      270

Thr Leu Pro Pro Ser Arg Asp Glu Leu Thr Lys Asn Gln Val Ser Leu
                      275                      280                      285

Trp Cys Leu Val Lys Gly Phe Tyr Pro Ser Asp Ile Ala Val Glu Trp
                      290                      295                      300

Glu Ser Asn Gly Gln Pro Glu Asn Asn Tyr Lys Thr Thr Pro Pro Val
 305                      310                      315                      320

Leu Asp Ser Asp Gly Ser Phe Phe Leu Tyr Ser Lys Leu Thr Val Asp
                      325                      330                      335

Lys Ser Arg Trp Gln Gln Gly Asn Val Phe Ser Cys Ser Val Met His
                      340                      345                      350

Glu Ala Leu His Asn His Tyr Thr Gln Lys Ser Leu Ser Leu Ser Pro
                      355                      360                      365

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Gly

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<210> SEQ ID NO 9
<211> LENGTH: 360
<212> TYPE: PRT
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Heavy chain amino acid sequence

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<400> SEQUENCE: 9

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Glu Val Gln Leu Gln Gln Ser Gly Pro Glu Leu Val Lys Pro Gly Ala
 1                      5                      10                      15

Ser Val Arg Met Ser Cys Lys Ala Ser Gly Tyr Thr Phe Thr Asn Tyr
                      20                      25                      30

Val Met His Trp Val Lys Gln Lys Pro Gly Gln Gly Leu Glu Trp Ile
                      35                      40                      45

Gly Tyr Ile Asn Pro Tyr Asn Asp Gly Thr Lys Tyr Ser Glu Lys Phe
 50                      55                      60

Lys Gly Lys Ala Thr Leu Thr Ser Asp Lys Ser Ser Ile Thr Ala Tyr
 65                      70                      75                      80

Met Glu Leu Ser Ser Leu Thr Ser Glu Asp Ser Ala Val Tyr Tyr Cys
                      85                      90                      95

Ala Lys Gly Gly Leu Phe Thr Phe Asp Tyr Trp Gly Gln Gly Thr Thr
                      100                      105                      110

Leu Thr Val Ser Ser Leu Gly Gly Cys Ser Gly Gly Gly Gly Ser Ala
                      115                      120                      125

Ser Glu Pro Lys Ser Ser Asp Lys Thr His Thr Cys Pro Pro Cys Pro
                      130                      135                      140

Ala Pro Glu Leu Leu Gly Gly Pro Ser Val Phe Leu Phe Pro Pro Lys
 145                      150                      155                      160

Pro Lys Asp Thr Leu Met Ile Ser Arg Thr Pro Glu Val Thr Cys Val
                      165                      170                      175

Val Val Asp Val Ser His Glu Asp Pro Glu Val Lys Phe Asn Trp Tyr
                      180                      185                      190

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Val	Asp	Gly	Val	Glu	Val	His	Asn	Ala	Lys	Thr	Lys	Pro	Arg	Glu	Glu
	195						200					205			
Gln	Tyr	Asn	Ser	Thr	Tyr	Arg	Val	Val	Ser	Val	Leu	Thr	Val	Leu	His
	210					215					220				
Gln	Asp	Trp	Leu	Asn	Gly	Lys	Glu	Tyr	Lys	Cys	Lys	Val	Ser	Asn	Lys
	225				230					235					240
Ala	Leu	Pro	Ala	Pro	Ile	Glu	Lys	Thr	Ile	Ser	Lys	Ala	Lys	Gly	Gln
			245						250					255	
Pro	Arg	Glu	Pro	Gln	Val	Tyr	Thr	Leu	Pro	Pro	Ser	Arg	Asp	Glu	Leu
			260					265					270		
Thr	Lys	Asn	Gln	Val	Ser	Leu	Trp	Cys	Leu	Val	Lys	Gly	Phe	Tyr	Pro
		275					280					285			
Ser	Asp	Ile	Ala	Val	Glu	Trp	Glu	Ser	Asn	Gly	Gln	Pro	Glu	Asn	Asn
	290					295					300				
Tyr	Lys	Thr	Thr	Pro	Pro	Val	Leu	Asp	Ser	Asp	Gly	Ser	Phe	Phe	Leu
	305				310					315					320
Tyr	Ser	Lys	Leu	Thr	Val	Asp	Lys	Ser	Arg	Trp	Gln	Gln	Gly	Asn	Val
			325						330					335	
Phe	Ser	Cys	Ser	Val	Met	His	Glu	Ala	Leu	His	Asn	His	Tyr	Thr	Gln
			340					345					350		
Lys	Ser	Leu	Ser	Leu	Ser	Pro	Gly								
		355					360								

<210> SEQ ID NO 10
 <211> LENGTH: 349
 <212> TYPE: PRT
 <213> ORGANISM: Artificial sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Light chain amino acid sequence

<400> SEQUENCE: 10

Glu	Val	Gln	Leu	Val	Glu	Ser	Gly	Gly	Gly	Leu	Val	Gln	Pro	Gly	Gly
1				5						10				15	
Ser	Leu	Arg	Leu	Ser	Cys	Ala	Ala	Ser	Tyr	Phe	Pro	Phe	Ser	Asp	Ser
			20					25					30		
Trp	Ile	His	Trp	Val	Arg	Gln	Ala	Pro	Gly	Lys	Gly	Leu	Glu	Trp	Val
		35				40						45			
Ala	Trp	Ile	Ser	Pro	Tyr	Gly	Gly	Ser	Ser	Tyr	Tyr	Ala	Asp	Ser	Val
	50					55					60				
Gln	Asp	Arg	Phe	Thr	Ile	Ser	Ala	Asp	Thr	Ser	Lys	Asn	Thr	Ala	Tyr
	65			70					75					80	
Leu	Gln	Met	Asn	Ser	Leu	Arg	Ala	Glu	Asp	Thr	Ala	Val	Tyr	Tyr	Cys
			85						90					95	
Ala	Lys	Arg	His	Trp	Pro	Gly	Gly	Phe	Asp	His	Trp	Gly	Gln	Gly	Thr
			100					105					110		
Leu	Val	Thr	Val	Ser	Ser	Gly	Gly	Gly	Gly	Ser	Gly	Gly	Gly	Gly	Ser
		115				120						125			
Gly	Gly	Gly	Gly	Ser	Ala	Ser	Asp	Ile	Gln	Met	Thr	Gln	Ser	Pro	Ser
	130					135					140				
Ser	Leu	Ser	Ala	Ser	Val	Gly	Asp	Arg	Val	Thr	Ile	Thr	Cys	Ser	Ala
	145				150					155					160
Ser	Gln	Asp	Val	Thr	Thr	Ala	Val	Ala	Trp	Tyr	Gln	Gln	Lys	Pro	Gly
			165						170					175	
Lys	Ala	Pro	Lys	Leu	Leu	Ile	Tyr	Ser	Ala	Ser	Phe	Pro	Tyr	Arg	Gly
			180					185						190	

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Val	Pro	Ser	Arg	Phe	Ser	Gly	Ser	Gly	Ser	Gly	Thr	Asp	Phe	Thr	Leu
	195						200					205			
Thr	Ile	Ser	Ser	Leu	Gln	Pro	Glu	Asp	Phe	Ala	Thr	Tyr	Tyr	Cys	Gln
	210					215					220				
Gln	Tyr	Met	Tyr	His	Pro	Ser	Thr	Phe	Gly	Gln	Gly	Thr	Lys	Val	Glu
	225				230					235					240
Ile	Lys	Arg	Thr	Val	Ala	Ala	Pro	Ser	Val	Phe	Ile	Phe	Pro	Pro	Ser
			245						250					255	
Asp	Glu	Gln	Leu	Lys	Ser	Gly	Thr	Ala	Ser	Val	Val	Cys	Leu	Leu	Asn
			260					265					270		
Asn	Phe	Tyr	Pro	Arg	Glu	Ala	Lys	Val	Gln	Trp	Lys	Val	Asp	Asn	Ala
	275						280					285			
Leu	Gln	Ser	Gly	Asn	Ser	Gln	Glu	Ser	Val	Thr	Glu	Gln	Asp	Ser	Lys
	290					295					300				
Asp	Ser	Thr	Tyr	Ser	Leu	Ser	Ser	Thr	Leu	Thr	Leu	Ser	Lys	Ala	Asp
	305				310					315					320
Tyr	Glu	Lys	His	Lys	Val	Tyr	Ala	Cys	Glu	Val	Thr	His	Gln	Gly	Leu
			325						330					335	
Ser	Ser	Pro	Val	Thr	Lys	Ser	Phe	Asn	Arg	Gly	Glu	Cys			
			340					345							

<210> SEQ ID NO 11
 <211> LENGTH: 353
 <212> TYPE: PRT
 <213> ORGANISM: Artificial sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Light chain amino acid sequence

<400> SEQUENCE: 11

Gln	Val	Gln	Leu	Val	Gln	Ser	Gly	Ala	Glu	Val	Lys	Lys	Pro	Gly	Ala
1			5						10					15	
Ser	Val	Lys	Val	Ser	Cys	Lys	Ala	Ser	Gly	Tyr	Thr	Phe	Thr	Asn	Tyr
		20						25					30		
Val	Leu	His	Trp	Val	Arg	Gln	Ala	Pro	Gly	Gln	Arg	Leu	Glu	Trp	Met
		35					40					45			
Gly	Tyr	Phe	Asn	Pro	Tyr	Asn	Asp	Asp	Ser	Lys	Tyr	Asn	Glu	Lys	Phe
	50					55					60				
Lys	Gly	Arg	Val	Thr	Ile	Thr	Arg	Asp	Thr	Ser	Ala	Ser	Thr	Ala	Tyr
	65				70				75						80
Met	Glu	Leu	Ser	Ser	Leu	Arg	Ser	Glu	Asp	Thr	Ala	Val	Tyr	Tyr	Cys
			85					90					95		
Ala	Arg	Gly	Gly	Tyr	Tyr	Ser	Met	Asp	Tyr	Trp	Gly	Gln	Gly	Thr	Leu
		100						105					110		
Val	Thr	Val	Ser	Ser	Gly	Gly	Gly	Gly	Ser	Gly	Gly	Gly	Gly	Ser	Gly
		115				120						125			
Gly	Gly	Gly	Ser	Ala	Ser	Asp	Ile	Val	Met	Thr	Gln	Thr	Pro	Leu	Ser
	130					135					140				
Leu	Pro	Val	Thr	Pro	Gly	Gln	Pro	Ala	Ser	Ile	Ser	Cys	Arg	Ala	Ser
	145				150					155					160
Gln	Ser	Leu	Ala	His	Ser	Asn	Gly	Tyr	Thr	Tyr	Leu	Gln	Trp	Tyr	Leu
			165					170						175	
Gln	Lys	Pro	Gly	Gln	Ser	Pro	Gln	Leu	Leu	Ile	Tyr	Lys	Val	Ser	Lys
		180						185					190		
Arg	Phe	Ser	Gly	Val	Pro	Asp	Arg	Phe	Ser	Gly	Ser	Gly	Ser	Gly	Thr
	195						200					205			

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Asp Phe Thr Leu Lys Ile Ser Arg Val Glu Ala Glu Asp Val Gly Val
 210                215                220

Tyr Tyr Cys Ser Gln Ser Thr His Val Pro Tyr Thr Phe Gly Gln Gly
 225                230                235                240

Thr Lys Val Glu Ile Lys Arg Thr Val Ala Ala Pro Ser Val Phe Ile
                245                250                255

Phe Pro Pro Ser Asp Glu Gln Leu Lys Ser Gly Thr Ala Ser Val Val
                260                265                270

Cys Leu Leu Asn Asn Phe Tyr Pro Arg Glu Ala Lys Val Gln Trp Lys
 275                280                285

Val Asp Asn Ala Leu Gln Ser Gly Asn Ser Gln Glu Ser Val Thr Glu
 290                295                300

Gln Asp Ser Lys Asp Ser Thr Tyr Ser Leu Ser Ser Thr Leu Thr Leu
 305                310                315                320

Ser Lys Ala Asp Tyr Glu Lys His Lys Val Tyr Ala Cys Glu Val Thr
                325                330                335

His Gln Gly Leu Ser Ser Pro Val Thr Lys Ser Phe Asn Arg Gly Glu
 340                345                350

Cys

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<210> SEQ ID NO 12
<211> LENGTH: 353
<212> TYPE: PRT
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Light chain amino acid sequence

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<400> SEQUENCE: 12

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Arg Thr Val Ala Ala Pro Ser Val Phe Ile Phe Pro Pro Ser Asp Glu
 1                5                10                15

Gln Leu Lys Ser Gly Thr Ala Ser Val Val Cys Leu Leu Asn Asn Phe
 20                25                30

Tyr Pro Arg Glu Ala Lys Val Gln Trp Lys Val Asp Asn Ala Leu Gln
 35                40                45

Ser Gly Asn Ser Gln Glu Ser Val Thr Glu Gln Asp Ser Lys Asp Ser
 50                55                60

Thr Tyr Ser Leu Ser Ser Thr Leu Thr Leu Ser Lys Ala Asp Tyr Glu
 65                70                75                80

Lys His Lys Val Tyr Ala Cys Glu Val Thr His Gln Gly Leu Ser Ser
 85                90                95

Pro Val Thr Lys Ser Phe Asn Arg Gly Glu Cys Gln Val Gln Leu Val
 100               105               110

Gln Ser Gly Ala Glu Val Lys Lys Pro Gly Ala Ser Val Lys Val Ser
 115               120               125

Cys Lys Ala Ser Gly Tyr Thr Phe Thr Asn Tyr Val Leu His Trp Val
 130               135               140

Arg Gln Ala Pro Gly Gln Arg Leu Glu Trp Met Gly Tyr Phe Asn Pro
 145               150               155               160

Tyr Asn Asp Asp Ser Lys Tyr Asn Glu Lys Phe Lys Gly Arg Val Thr
 165               170               175

Ile Thr Arg Asp Thr Ser Ala Ser Thr Ala Tyr Met Glu Leu Ser Ser
 180               185               190

Leu Arg Ser Glu Asp Thr Ala Val Tyr Tyr Cys Ala Arg Gly Gly Tyr
 195               200               205

Tyr Ser Met Asp Tyr Trp Gly Gln Gly Thr Leu Val Thr Val Ser Ser

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210	215	220
Gly Gly Gly Gly Ser Gly Gly Gly Gly Ser Gly Gly Gly Gly Ser Ala		
225	230	235 240
Ser Asp Ile Val Met Thr Gln Thr Pro Leu Ser Leu Pro Val Thr Pro		
245	250	255
Gly Gln Pro Ala Ser Ile Ser Cys Arg Ala Ser Gln Ser Leu Ala His		
260	265	270
Ser Asn Gly Tyr Thr Tyr Leu Gln Trp Tyr Leu Gln Lys Pro Gly Gln		
275	280	285
Ser Pro Gln Leu Leu Ile Tyr Lys Val Ser Lys Arg Phe Ser Gly Val		
290	295	300
Pro Asp Arg Phe Ser Gly Ser Gly Ser Gly Thr Asp Phe Thr Leu Lys		
305	310	315 320
Ile Ser Arg Val Glu Ala Glu Asp Val Gly Val Tyr Tyr Cys Ser Gln		
325	330	335
Ser Thr His Val Pro Tyr Thr Phe Gly Gln Gly Thr Lys Val Glu Ile		
340	345	350

Lys

<210> SEQ ID NO 13
 <211> LENGTH: 349
 <212> TYPE: PRT
 <213> ORGANISM: Artificial sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Light chain amino acid sequence

<400> SEQUENCE: 13

Arg Thr Val Ala Ala Pro Ser Val Phe Ile Phe Pro Pro Ser Asp Glu		
1	5	10 15
Gln Leu Lys Ser Gly Thr Ala Ser Val Val Cys Leu Leu Asn Asn Phe		
20	25	30
Tyr Pro Arg Glu Ala Lys Val Gln Trp Lys Val Asp Asn Ala Leu Gln		
35	40	45
Ser Gly Asn Ser Gln Glu Ser Val Thr Glu Gln Asp Ser Lys Asp Ser		
50	55	60
Thr Tyr Ser Leu Ser Ser Thr Leu Thr Leu Ser Lys Ala Asp Tyr Glu		
65	70	75 80
Lys His Lys Val Tyr Ala Cys Glu Val Thr His Gln Gly Leu Ser Ser		
85	90	95
Pro Val Thr Lys Ser Phe Asn Arg Gly Glu Cys Glu Val Gln Leu Val		
100	105	110
Glu Ser Gly Gly Gly Leu Val Gln Pro Gly Gly Ser Leu Arg Leu Ser		
115	120	125
Cys Ala Ala Ser Tyr Phe Pro Phe Ser Asp Ser Trp Ile His Trp Val		
130	135	140
Arg Gln Ala Pro Gly Lys Gly Leu Glu Trp Val Ala Trp Ile Ser Pro		
145	150	155 160
Tyr Gly Gly Ser Ser Tyr Tyr Ala Asp Ser Val Gln Asp Arg Phe Thr		
165	170	175
Ile Ser Ala Asp Thr Ser Lys Asn Thr Ala Tyr Leu Gln Met Asn Ser		
180	185	190
Leu Arg Ala Glu Asp Thr Ala Val Tyr Tyr Cys Ala Lys Arg His Trp		
195	200	205
Pro Gly Gly Phe Asp His Trp Gly Gln Gly Thr Leu Val Thr Val Ser		
210	215	220

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Ser Gly Gly Gly Gly Ser Gly Gly Gly Gly Ser Gly Gly Gly Gly Ser
 225 230 235 240
 Ala Ser Asp Ile Gln Met Thr Gln Ser Pro Ser Ser Leu Ser Ala Ser
 245 250 255
 Val Gly Asp Arg Val Thr Ile Thr Cys Ser Ala Ser Gln Asp Val Thr
 260 265 270
 Thr Ala Val Ala Trp Tyr Gln Gln Lys Pro Gly Lys Ala Pro Lys Leu
 275 280 285
 Leu Ile Tyr Ser Ala Ser Phe Pro Tyr Arg Gly Val Pro Ser Arg Phe
 290 295 300
 Ser Gly Ser Gly Ser Gly Thr Asp Phe Thr Leu Thr Ile Ser Ser Leu
 305 310 315 320
 Gln Pro Glu Asp Phe Ala Thr Tyr Tyr Cys Gln Gln Tyr Met Tyr His
 325 330 335
 Pro Ser Thr Phe Gly Gln Gly Thr Lys Val Glu Ile Lys
 340 345

<210> SEQ ID NO 14
 <211> LENGTH: 214
 <212> TYPE: PRT
 <213> ORGANISM: Artificial sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Light chain amino acid sequence

<400> SEQUENCE: 14

Asp Ile Gln Met Thr Gln Ser Pro Ser Ser Leu Ser Ala Ser Val Gly
 1 5 10 15
 Asp Arg Val Thr Ile Thr Cys Ser Ala Ser Gln Asp Val Thr Thr Ala
 20 25 30
 Val Ala Trp Tyr Gln Gln Lys Pro Gly Lys Ala Pro Lys Leu Leu Ile
 35 40 45
 Tyr Ser Ala Ser Phe Pro Tyr Arg Gly Val Pro Ser Arg Phe Ser Gly
 50 55 60
 Ser Gly Ser Gly Thr Asp Phe Thr Leu Thr Ile Ser Ser Leu Gln Pro
 65 70 75 80
 Glu Asp Phe Ala Thr Tyr Tyr Cys Gln Gln Tyr Met Tyr His Pro Ser
 85 90 95
 Thr Phe Gly Gln Gly Thr Lys Val Glu Ile Lys Arg Thr Val Ala Ala
 100 105 110
 Pro Ser Val Phe Ile Phe Pro Pro Ser Asp Glu Gln Leu Lys Ser Gly
 115 120 125
 Thr Ala Ser Val Val Cys Leu Leu Asn Asn Phe Tyr Pro Arg Glu Ala
 130 135 140
 Lys Val Gln Trp Lys Val Asp Asn Ala Leu Gln Ser Gly Asn Ser Gln
 145 150 155 160
 Glu Ser Val Thr Glu Gln Asp Ser Lys Asp Ser Thr Tyr Ser Leu Ser
 165 170 175
 Ser Thr Leu Thr Leu Ser Lys Ala Asp Tyr Glu Lys His Lys Val Tyr
 180 185 190
 Ala Cys Glu Val Thr His Gln Gly Leu Ser Ser Pro Val Thr Lys Ser
 195 200 205
 Phe Asn Arg Gly Glu Cys
 210

<210> SEQ ID NO 15

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<211> LENGTH: 116
<212> TYPE: PRT
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Light chain amino acid sequence

<400> SEQUENCE: 15
Asp Val Val Met Thr Gln Thr Pro Leu Ser Leu Pro Val Ser Leu Gly
1          5          10          15
Asp Gln Ala Ser Ile Ser Cys Arg Ser Ser Gln Ser Leu Val His Ser
20          25          30
Lys Gly Asn Thr Tyr Leu His Trp Tyr Leu Gln Lys Pro Gly Gln Ser
35          40          45
Pro Lys Leu Leu Ile Tyr Arg Val Ser Asn Arg Phe Ser Gly Val Pro
50          55          60
Asp Arg Phe Ser Gly Ser Gly Ser Gly Thr Asp Phe Thr Leu Lys Ile
65          70          75          80
Ser Arg Val Glu Ala Glu Asp Leu Gly Val Tyr Phe Cys Ser Gln Ser
85          90          95
Thr His Val Pro Phe Thr Phe Gly Ser Gly Thr Lys Leu Glu Ile Lys
100         105         110
Leu Gly Gly Cys
115

<210> SEQ ID NO 16
<211> LENGTH: 124
<212> TYPE: PRT
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Light chain amino acid sequence

<400> SEQUENCE: 16
Asp Val Val Met Thr Gln Thr Pro Leu Ser Leu Pro Val Ser Leu Gly
1          5          10          15
Asp Gln Ala Ser Ile Ser Cys Arg Ser Ser Gln Ser Leu Val His Ser
20          25          30
Lys Gly Asn Thr Tyr Leu His Trp Tyr Leu Gln Lys Pro Gly Gln Ser
35          40          45
Pro Lys Leu Leu Ile Tyr Arg Val Ser Asn Arg Phe Ser Gly Val Pro
50          55          60
Asp Arg Phe Ser Gly Ser Gly Ser Gly Thr Asp Phe Thr Leu Lys Ile
65          70          75          80
Ser Arg Val Glu Ala Glu Asp Leu Gly Val Tyr Phe Cys Ser Gln Ser
85          90          95
Thr His Val Pro Phe Thr Phe Gly Ser Gly Thr Lys Leu Glu Ile Lys
100         105         110
Ser Gly Gly Gly Gly Ser Ala Ser Leu Gly Gly Cys
115         120

<210> SEQ ID NO 17
<211> LENGTH: 133
<212> TYPE: PRT
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: Light chain amino acid sequence

<400> SEQUENCE: 17
Asp Val Val Met Thr Gln Thr Pro Leu Ser Leu Pro Val Ser Leu Gly
1          5          10          15

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Asp Gln Ala Ser Ile Ser Cys Arg Ser Ser Gln Ser Leu Val His Ser
 20 25 30
 Lys Gly Asn Thr Tyr Leu His Trp Tyr Leu Gln Lys Pro Gly Gln Ser
 35 40 45
 Pro Lys Leu Leu Ile Tyr Arg Val Ser Asn Arg Phe Ser Gly Val Pro
 50 55 60
 Asp Arg Phe Ser Gly Ser Gly Ser Gly Thr Asp Phe Thr Leu Lys Ile
 65 70 75 80
 Ser Arg Val Glu Ala Glu Asp Leu Gly Val Tyr Phe Cys Ser Gln Ser
 85 90 95
 Thr His Val Pro Phe Thr Phe Gly Ser Gly Thr Lys Leu Glu Ile Lys
 100 105 110
 Gly Gly Gly Gly Ser Gly Gly Gly Gly Ser Gly Gly Gly Gly Ser Ala
 115 120 125
 Ser Leu Gly Gly Cys
 130

<210> SEQ ID NO 18
 <211> LENGTH: 112
 <212> TYPE: PRT
 <213> ORGANISM: Artificial sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: VL

<400> SEQUENCE: 18

Asp Ile Val Met Thr Gln Thr Pro Leu Ser Leu Pro Val Thr Pro Gly
 1 5 10 15
 Gln Pro Ala Ser Ile Ser Cys Arg Ala Ser Gln Ser Leu Ala His Ser
 20 25 30
 Asn Gly Tyr Thr Tyr Leu Gln Trp Tyr Leu Gln Lys Pro Gly Gln Ser
 35 40 45
 Pro Gln Leu Leu Ile Tyr Lys Val Ser Lys Arg Phe Ser Gly Val Pro
 50 55 60
 Asp Arg Phe Ser Gly Ser Gly Ser Gly Thr Asp Phe Thr Leu Lys Ile
 65 70 75 80
 Ser Arg Val Glu Ala Glu Asp Val Gly Val Tyr Tyr Cys Ser Gln Ser
 85 90 95
 Thr His Val Pro Tyr Thr Phe Gly Gln Gly Thr Lys Val Glu Ile Lys
 100 105 110

<210> SEQ ID NO 19
 <211> LENGTH: 117
 <212> TYPE: PRT
 <213> ORGANISM: Artificial sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: VH

<400> SEQUENCE: 19

Gln Val Gln Leu Val Gln Ser Gly Ala Glu Val Lys Lys Pro Gly Ala
 1 5 10 15
 Ser Val Lys Val Ser Cys Lys Ala Ser Gly Tyr Thr Phe Thr Asn Tyr
 20 25 30
 Val Leu His Trp Val Arg Gln Ala Pro Gly Gln Arg Leu Glu Trp Met
 35 40 45
 Gly Tyr Phe Asn Pro Tyr Asn Asp Asp Ser Lys Tyr Asn Glu Lys Phe
 50 55 60
 Lys Gly Arg Val Thr Ile Thr Arg Asp Thr Ser Ala Ser Thr Ala Tyr
 65 70 75 80

-continued

Met Glu Leu Ser Ser Leu Arg Ser Glu Asp Thr Ala Val Tyr Tyr Cys
85 90 95

Ala Arg Gly Gly Tyr Tyr Ser Met Asp Tyr Trp Gly Gln Gly Thr Leu
100 105 110

Val Thr Val Ser Ser
115

<210> SEQ ID NO 20
<211> LENGTH: 107
<212> TYPE: PRT
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: VL

<400> SEQUENCE: 20

Asp Ile Gln Met Thr Gln Ser Pro Ser Ser Leu Ser Ala Ser Val Gly
1 5 10 15

Asp Arg Val Thr Ile Thr Cys Ser Ala Ser Gln Asp Val Thr Thr Ala
20 25 30

Val Ala Trp Tyr Gln Gln Lys Pro Gly Lys Ala Pro Lys Leu Leu Ile
35 40 45

Tyr Ser Ala Ser Phe Pro Tyr Arg Gly Val Pro Ser Arg Phe Ser Gly
50 55 60

Ser Gly Ser Gly Thr Asp Phe Thr Leu Thr Ile Ser Ser Leu Gln Pro
65 70 75 80

Glu Asp Phe Ala Thr Tyr Tyr Cys Gln Gln Tyr Met Tyr His Pro Ser
85 90 95

Thr Phe Gly Gln Gly Thr Lys Val Glu Ile Lys
100 105

<210> SEQ ID NO 21
<211> LENGTH: 118
<212> TYPE: PRT
<213> ORGANISM: Artificial sequence
<220> FEATURE:
<223> OTHER INFORMATION: VH

<400> SEQUENCE: 21

Glu Val Gln Leu Val Glu Ser Gly Gly Gly Leu Val Gln Pro Gly Gly
1 5 10 15

Ser Leu Arg Leu Ser Cys Ala Ala Ser Tyr Phe Pro Phe Ser Asp Ser
20 25 30

Trp Ile His Trp Val Arg Gln Ala Pro Gly Lys Gly Leu Glu Trp Val
35 40 45

Ala Trp Ile Ser Pro Tyr Gly Gly Ser Ser Tyr Tyr Ala Asp Ser Val
50 55 60

Gln Asp Arg Phe Thr Ile Ser Ala Asp Thr Ser Lys Asn Thr Ala Tyr
65 70 75 80

Leu Gln Met Asn Ser Leu Arg Ala Glu Asp Thr Ala Val Tyr Tyr Cys
85 90 95

Ala Lys Arg His Trp Pro Gly Gly Phe Asp His Trp Gly Gln Gly Thr
100 105 110

Leu Val Thr Val Ser Ser
115

<210> SEQ ID NO 22
<211> LENGTH: 454
<212> TYPE: PRT

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<213> ORGANISM: Artificial sequence

<220> FEATURE:

<223> OTHER INFORMATION: Human PD-L1-mFc

<400> SEQUENCE: 22

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Phe Thr Val Thr Val Pro Lys Asp Leu Tyr Val Val Glu Tyr Gly Ser
 1              5              10              15

Asn Met Thr Ile Glu Cys Lys Phe Pro Val Glu Lys Gln Leu Asp Leu
      20              25              30

Ala Ala Leu Ile Val Tyr Trp Glu Met Glu Asp Lys Asn Ile Ile Gln
      35              40              45

Phe Val His Gly Glu Glu Asp Leu Lys Val Gln His Ser Ser Tyr Arg
      50              55              60

Gln Arg Ala Arg Leu Leu Lys Asp Gln Leu Ser Leu Gly Asn Ala Ala
      65              70              75              80

Leu Gln Ile Thr Asp Val Lys Leu Gln Asp Ala Gly Val Tyr Arg Cys
      85              90              95

Met Ile Ser Tyr Gly Gly Ala Asp Tyr Lys Arg Ile Thr Val Lys Val
      100             105             110

Asn Ala Pro Tyr Asn Lys Ile Asn Gln Arg Ile Leu Val Val Asp Pro
      115             120             125

Val Thr Ser Glu His Glu Leu Thr Cys Gln Ala Glu Gly Tyr Pro Lys
      130             135             140

Ala Glu Val Ile Trp Thr Ser Ser Asp His Gln Val Leu Ser Gly Lys
      145             150             155             160

Thr Thr Thr Thr Asn Ser Lys Arg Glu Glu Lys Leu Phe Asn Val Thr
      165             170             175

Ser Thr Leu Arg Ile Asn Thr Thr Thr Asn Glu Ile Phe Tyr Cys Thr
      180             185             190

Phe Arg Arg Leu Asp Pro Glu Glu Asn His Thr Ala Glu Leu Val Ile
      195             200             205

Pro Glu Leu Pro Leu Ala His Pro Pro Asn Glu Arg Thr Ser Gly Gly
      210             215             220

Gly Gly Ser Val Pro Arg Asp Cys Gly Cys Lys Pro Cys Ile Cys Thr
      225             230             235             240

Val Pro Glu Val Ser Ser Val Phe Ile Phe Pro Pro Lys Pro Lys Asp
      245             250             255

Val Leu Thr Ile Thr Leu Thr Pro Lys Val Thr Cys Val Val Val Asp
      260             265             270

Ile Ser Lys Asp Asp Pro Glu Val Gln Phe Ser Trp Phe Val Asp Asp
      275             280             285

Val Glu Val His Thr Ala Gln Thr Gln Pro Arg Glu Glu Gln Phe Asn
      290             295             300

Ser Thr Phe Arg Ser Val Ser Glu Leu Pro Ile Met His Gln Asp Trp
      305             310             315             320

Leu Asn Gly Lys Glu Phe Lys Cys Arg Val Asn Ser Ala Ala Phe Pro
      325             330             335

Ala Pro Ile Glu Lys Thr Ile Ser Lys Thr Lys Gly Arg Pro Lys Ala
      340             345             350

Pro Gln Val Tyr Thr Ile Pro Pro Pro Lys Glu Gln Met Ala Lys Asp
      355             360             365

Lys Val Ser Leu Thr Cys Met Ile Thr Asp Phe Phe Pro Glu Asp Ile
      370             375             380

Thr Val Glu Trp Gln Trp Asn Gly Gln Pro Ala Glu Asn Tyr Lys Asn

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385	390	395	400
Thr Gln Pro Ile Met Asp Thr Asp Gly Ser Tyr Phe Val Tyr Ser Lys			
	405	410	415
Leu Asn Val Gln Lys Ser Asn Trp Glu Ala Gly Asn Thr Phe Thr Cys			
	420	425	430
Ser Val Leu His Glu Gly Leu His Asn His His Thr Glu Lys Ser Leu			
	435	440	445
Ser His Ser Pro Gly Lys			
450			

<210> SEQ ID NO 23
 <211> LENGTH: 134
 <212> TYPE: PRT
 <213> ORGANISM: Artificial sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Human CD47-His

<400> SEQUENCE: 23

Gln Leu Leu Phe Asn Lys Thr Lys Ser Val Glu Phe Thr Phe Cys Asn			
1	5	10	15
Asp Thr Val Val Ile Pro Cys Phe Val Thr Asn Met Glu Ala Gln Asn			
	20	25	30
Thr Thr Glu Val Tyr Val Lys Trp Lys Phe Lys Gly Arg Asp Ile Tyr			
	35	40	45
Thr Phe Asp Gly Ala Leu Asn Lys Ser Thr Val Pro Thr Asp Phe Ser			
	50	55	60
Ser Ala Lys Ile Glu Val Ser Gln Leu Leu Lys Gly Asp Ala Ser Leu			
	65	70	75
Lys Met Asp Lys Ser Asp Ala Val Ser His Thr Gly Asn Tyr Thr Cys			
	85	90	95
Glu Val Thr Glu Leu Thr Arg Glu Gly Glu Thr Ile Ile Glu Leu Lys			
	100	105	110
Tyr Arg Val Val Ser Trp Phe Ser Pro Ser Gly Gly Gly Gly Ser His			
	115	120	125
His His His His His His			
130			

<210> SEQ ID NO 24
 <211> LENGTH: 7
 <212> TYPE: PRT
 <213> ORGANISM: Artificial sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: IgG1 partial hinge region

<400> SEQUENCE: 24

Glu Pro Lys Ser Cys Asp Lys
1 5

<210> SEQ ID NO 25
 <211> LENGTH: 5
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Flexible peptide

<400> SEQUENCE: 25

Gly Gly Gly Gly Ser
1 5

<210> SEQ ID NO 26

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<211> LENGTH: 7
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Flexible peptide

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<400> SEQUENCE: 26

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Gly Gly Gly Gly Ser Ala Ser
1             5

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<210> SEQ ID NO 27
<211> LENGTH: 4
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Linker

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<400> SEQUENCE: 27

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Leu Gly Gly Cys
1

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<210> SEQ ID NO 28
<211> LENGTH: 4
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Linker

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<400> SEQUENCE: 28

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Gly Gly Gly Cys
1

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<210> SEQ ID NO 29
<211> LENGTH: 15
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Flexible peptide

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<400> SEQUENCE: 29

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Gly Gly Gly Gly Ser Gly Gly Gly Gly Gly Gly Gly Gly Ser
1             5             10             15

```

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<210> SEQ ID NO 30
<211> LENGTH: 21
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Flexible peptide

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<400> SEQUENCE: 30

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Gly Gly Gly Gly Ser Ala Ser Gly Gly Gly Gly Ser Ala Ser Gly Gly
1             5             10             15

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Gly Gly Ser Ala Ser
20

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The invention claimed is:

1. A bivalent bispecific antibody, comprising

- a1) scFv2 that is a single-chain variable fragment (scFv) derived from an antibody that specifically binds to a first antigen, a flexible peptide, a heavy chain constant region CH1 and a partial hinge region of IgG1 in an order of CH1-partial hinge-flexible peptide scFv2, and
- b1) scFv1 that is a single-chain variable fragment (scFv) derived from an antibody that specifically binds to a second antigen and a light chain constant region CL in an order of scFv1-CL; or

a2) scFv2 that is a single-chain variable fragment specifically binding to a first antigen, a flexible peptide, a heavy chain constant region CH1 and a partial hinge region of IgG1 in an order of scFv2-flexible peptide-CH1-partial hinge, and

b2) scFv1 that is a single-chain variable fragment specifically binding to a second antigen and a light chain constant region CL in an order of CL-scFv1; wherein the order is from N-terminal to C-terminal.

2. The bivalent bispecific antibody according to claim 1, wherein the flexible peptide comprises a sequence of

57

(GGGGS (SEQ ID NO: 25))_n or (GGGGSAS (SEQ ID NO: 26))_n, wherein n is an integer greater than 0, and the IgG1 partial hinge region linked to the flexible peptide comprises a sequence of EPKSCDK (SEQ ID NO: 24).

3. The bivalent bispecific antibody according to claim 1, wherein CL and CH1 form a heterodimer via a disulfide bond formed between the terminal cysteine residue in CL and the cysteine residue in the heavy chain hinge region.

4. The bivalent bispecific antibody according to claim 1, where in the bivalent bispecific antibody comprises two chains,

1) one chain has an amino acid sequence as shown in SEQ ID NO: 2, and

the other chain has an amino acid sequence as shown in SEQ ID NO: 11;

2) one chain has an amino acid sequence as shown in SEQ ID NO: 1, and the other chain has an amino acid sequence as shown in SEQ ID NO: 10;

3) one chain has an amino acid sequence as shown in SEQ ID NO: 3, and the other chain has an amino acid sequence as shown in SEQ ID NO: 12; or

4) one chain has an amino acid sequence as shown in SEQ ID NO: 4, and the other chain has an amino acid sequence as shown in SEQ ID NO: 13.

58

5. A method for producing the bivalent bispecific antibody according to claim 1, comprising

a) transforming a host cell with

a first vector, comprising a nucleic acid encoding a scFv specifically binding to a first antigen, a flexible peptide, a heavy chain constant region CH1 and a partial hinge region of IgG1, and

a second vector, comprising a nucleic acid encoding a scFv specifically binding to a second antigen and a light chain constant region CL;

b) culturing the host cell under conditions that allow the synthesis of an antibody; and

c) recovering the antibody from the culture.

6. A nucleic acid molecule encoding the bivalent bispecific antibody of claim 1.

7. A host cell, comprising a nucleic acid molecule encoding the bivalent bispecific antibody of claim 1.

8. A composition comprising the bivalent bispecific antibody of claim 1, wherein the composition is a pharmaceutical composition or a diagnostic composition.

* * * * *